Module description Catalog Program in Applied Mathematics

FACULTY OF MATHEMATICS AND COMPUTER SCIENCE
UNIVERSITY OF SCIENCE
VIETNAM NATIONAL UNIVERISTY HO CHI MINH CITY

Contents

I. GENERAL EDUCATION 1
Basic Economics - BAA00005 1
Earth Science - GEO00002 3
English 1 - BAA00011 4
English 2 - BAA00012 6
English 3 - BAA00013 8
English 4 - BAA00014 9
Environmental Studies - ENV00001 11
General Biology I - BIO00001 13
General Biology II - BIO00002 15
General Chemistry 1 - CHE00001 17
General Chemistry 2 - CHE00002 18
General Chemistry Laboratory 1 - CHE00081 20
General Chemistry Laboratory 2 - CHE00082 21
General Law - BAA00004 22
General Physics 1 (Mechanics and Thermodynamics) - PHY00001 24
General Physics 2 (Electromagnetism - Optics) - PHY00002 25
History of Vietnamese Communist Party - BAA00104 27
Ho Chi Minh's Ideology - BAA00003 29
Human and Environment - ENV00003 31
Introduction to Informatics - CSC00003 33
Labwork on General Biology I - BIO00081 35
Labwork on General Biology II - BIO00082 37
General Physics Lab - PHY00081 39
Marxist-Leninist Philosophy - BAA00101 41
Marxist-Leninist Political Economy - BAA00102 43
Psychology - BAA00006 45
Scientific Socialism - BAA00103 47
Teamwork and Learning Skills - BAA00008 49
Analysis 1A - MTH00010 50
Analysis 2A - MTH00012 51
Analysis 3A - MTH00014 52

Ordinary Differential Equations - MTH00015 54
Calculus 1A - MTH00011 55
Calculus 2A - MTH00013 56
Computational Software Laboratory - MTH00087 57
Linear Algebra - MTH00030 58
Linear Algebra Practice - MTH00083 60
Abstract Algebra - MTH00031 61
Abstract Algebra Practice - MTH00084 62
Introduction to Computer Programming - MTH00055 64
II. PROFESSIONAL EDUCATION KNOWLEDGE 66
1. BASIC KNOWLEDGE 66
Algebra 2A - MTH10402 66
Functional Analysis - MTH10403 67
Measure Theory and Probability - MTH10401 69
Mathematical Statistics - MTH10404 70
2. SPECIALIZATION IN MECHANICS 73
2.1. COMPULSORY COURSES 73
Numerical Analysis 1 - MTH10410 73
Equations of Mathematical Physics - MTH10413 74
Theoretical Mechanics - MTH10427 75
Continuum Mechanics - MTH10428 77
Finite Element Method - MTH10429 78
Complex variable functions - MTH10412 80
Solids Mechanics - MTH10434 81
Fluid Mechanics - MTH10435 83
2.2. ELECTIVE COURSES 85
Symbolic Programming for Applied Problems - MTH10611 85
Mechanics Seminar - MTH10520 86
3. SPECIALIZATION IN DIDACTICS AND METHODOLOGY OF TEACHING MATHEMATICS 89
3.1. COMPULSORY COURSES 89
Pedagogical Psychology - MTH10101 89
Optimal Teaching and Learning Methods - MTH10102 91

General Education - MTH10104 93
Didactics of Teaching - MTH10105 95
Elementary Number Theory and Mathematical Logic - MTH10106 97
Methods of Teaching Mathematics 1- MTH10110 99
Methods of Teaching Mathematics 2- MTH10111 101
Pedagogical Practice - MTH10112 103
3.2. ELECTIVE COURSES 104
Quality and Quality Management - MTH10103 104
Teaching Mathematics in English 1 - MTH10119 106
Teaching Mathematics in English 2 - MTH10120 108
Elementary Algebra – MTH10121 110
Elementary Geometry - MTH10122 112
Application of Advanced Mathematical Methods in Solving Complex High School's Mathematics Problem - MTH10123 113
Classroom Management and Organization - MTH10124 115
Classroom Assessment Techniques – MTH10125 117
Pedagogy Seminar - MTH10126 119
4. SPECIALIZATION IN MATHEMATICAL FINANCE 122
4.1. COMPULSORY COURSES 122
Elementary Financial Mathematics - MTH10201 122
Forecasting - MTH10202 123
Mathematical Finance Models - MTH10203 125
Financial and Monetary Theory - MTH10204 127
Advanced Financial Mathematics - MTH10209 129
4.2. ELECTIVE COURSES 130
Quantitative Microeconomics - MTH10214 130
Corporate finance - MTH10217 131
Financial Analysis - MTH10219 133
Basic Actuarial Mathematics - MTH10220 135
Advanced Actuarial Mathematics - MTH10221 137
5. SPECIALIZATION IN OPTIMIZATION 139
5.1. COMPULSORY COURSES 139
Operations Research - MTH10446 139

Nonlinear programming - MTH10447 140
Optimization Models in Economics - MTH10615 142
Linear programming - MTH10449 143
Optimization Algorithms - MTH10450 144
Introduction to Convex Analysis and Convex Programming - MTH10543 146
5.2. ELECTIVE COURSES 147
Game Theory - MTH10541 147
Optimal Control - MTH10545 149
Multi-objective Optimization - MTH10553 150
Seminar on Optimization - MTH10616 151
6. GRADUATING WORKS 153
Internship - MTH10549 153
Graduation Thesis - MTH10595 154
Surveys of Mathematics, Computer Science, and specializations - MTH10617 155

I. GENERAL EDUCATION

Basic Economics - BAA00005

Module designation	Basic Economics
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Le Nhan My
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	Total workload: 90 hours. Contact hours (lecture): 30 hours. Private study including examination preparation, specified in hours: 60 hours.
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Knowledge: Forming and developing (one step) capacity to collect information, skills to synthesize and systematize issues in an overall relationship; skills to compare, analyze, comment, and evaluate micro-economic issues. Grasp the basic content of Microeconomics - a part of economics. Specifically: • Understand the theory of economic choice, the influence of the law of scarcity, and economic models on economic choice. • Understand the theory of supply and demand. • Understand the theory of consumer behavior. • Understand the theory of producer behavior. • Understand the theory of competition and monopoly. • Understand the theory of factor markets.

	• Understand the theory of the role of government. • Understand the analysis of the influence of factors on the balance of the market, in terms of skills. • Having the ability to apply the knowledge learned to study the nature of economic phenomena, the laws, and trends of the phenomena, and the laws of the market economy. • Ability to apply the knowledge learned in the study of macroeconomics, development economics, and several other economic subjects. Skills: • Forming and developing (one step) capacity to collect information, skills to synthesize and systematize issues in an overall relationship; skills to compare, analyze, comment, and evaluate micro-economic issues. • Develop reasoning and public speaking skills. Attitude: • Trying to be righteous in recognizing and evaluating the lines, policies, and laws of the State of Vietnam in the development of the market economy with the state's regulation. Other Objectives: • Through presentations and problem-solving. Forming and developing collaboration and teamwork skills. • Develop skills of creative thinking, discovery, and discovery. • Cultivate and develop assessment and self-assessment capacity. • Develop public speaking and commenting skills.
Content	The course presents some basic problems of economics; principles of economics, supply and demand patterns and market equilibrium; theory of consumer behavior and business behavior; types of markets; aggregate supply, aggregate demand, and measure national output. The module includes the following chapters: Chapter 1: Economics and Fundamental Issues Chapter 2: Markets, Supply and Demand, and Market Equilibrium

	Chapter 3: Theory of Consumer Behavior Chapter 4: Production Cost Theory and Profit Maximization Chapter 5: National Output Measurement, Inflation, and Unemployment
Examination forms	Homework: 20%. Midterm exam: 20%. Final exam: 60%.
Study and examination requirements	Regulations on time, attendance, and discipline in the course: attend class on time and at least 70% of the sessions (only to be absent for a maximum of 30%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have all test scores, lively discussions, constructive and serious statements in class.
Reading list	None

Earth Science - GEO00002

Module designation	Earth science
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at Faculty of Geology, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	Total: 60 hours. Contact hours: 30 lectures hours (in class) Private study: 30 hours (self-study)
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Earth Science introduces general knowledge about the Earth related to the rights of the Earth, inside and outside the Earth including position and operation of the earth in space;

	composition and structure of the atmosphere, wind, and weather, climate, climate change; distribution of freshwater in hydrosphere, sea and ocean behavior, El Nino-La Nina phenomena: Geosphere: composition and structure of the earth's crust, weathering, erosion-accumulation, landslides, the internal structure of the earth, earthquakes, volcanoes, plate tectonic activities; learn about the history of the earth through the record of fossil remains. Knowledge of Earth science is a necessary basis for understanding the natural environment of the Earth.
Content	1. Presentation on deformations of the earth's crust and earthquake, volcanic activities and membrane tectonic mechanism. 2. Learn about Earth's history through fossil and stratigraphic records. 3. Apply this knowledge to explain some issues in the main profession. 4. Skilled in group discussion, presentation and criticism Attitude, diligence: enthusiasm, honesty in learning; Serious and honest in checking.
Examination forms	Writing (Midterm: 30%, Final exam: 70%)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Earth Science*, DANIELSON, E.W., DENECKE. EJ. Ir. 1986. 2. *Foundations of Earth Science*, Lutgens Frederick K. Tarbuck Edward, 1, 1997. 3. *Earth Science Textbook*, LUU DUC HAI, TRAN NGHI. 2008.

English 1 - BAA00011

Module designation	English 1
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at Foreign language center of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory

Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading, and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as leisure activities, important life events, emotion, attitude, physical appearance description, travel plans, presenting dreams, countries, people, and languages. Learners can understand and use grammar structures at the pre-intermediate level such as basic tenses and other related matters. Learners will be able to choose the answer that best describes the given picture, choose the correct response to the questions, and understand dialogues and short monologues. Learners will be able to pronounce single words, word clusters and sentences, describe a given picture, and build basic communications in daily life. Learners will be able to comprehend 300-500 words passage of familiar topics and gain more knowledge of different cultures around the world. Learners can write essays about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English major students at the University of Science - Vietnam National University - Hochiminh City, using the first eight modules in the book New Cutting Edge (Pre-intermediate). These modules cover

	vocabulary, grammar, reading, listening, speaking, and writing in a wide range of topics such as leisure activities, important life events, feelings and emotions, attitudes, physical appearance descriptions, travel plans, presenting dreams, countries, people, and languages. Students need to complete various tasks, including presentations, debates, role-plays, assignments, tests and so on.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: workbook*. Harlow: Pearson Education.

English 2 - BAA00012

Module designation	English 2
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at Foreign Language Centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • English 1 (BAA00011)

Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, finance- related issues. Learners can understand and use grammar structures in pre-intermediate level such as basic tenses and more complex grammatical structures including conditional sentences, passive, and verb patterns. Learners will be able to choose the correct response for the questions and understand dialogues and short monologues. Learners will be able to pronounce words, generate short conversations, discuss real-life familiar topics, understand, and quickly respond to generated questions, and improve basic communication skills in daily life. Learners will be able to comprehend 500 - 700 words passage of familiar topics and gain more knowledge of different cultures around the world. Learners can write appropriate responses to written requests or complaints in business and social contexts, applying theories into real life practice.
Content	This course is designed for non-English major students at the University of Science - Vietnam National University - Ho Chi Minh City, using seven modules (modules 09-15) in the book New Cutting Edge (Pre-intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics such as everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, and finance-related issues. Students need to complete various tasks, including presentations, debates, role-plays, assignments, tests and so on.
Examination forms	Writing (Midterm: 30%, Final exam: 70%)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.

Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: workbook*. Harlow: Pearson Education.

English 3 - BAA00013

Module designation	English 3
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Foreign Language Centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours. Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • English 2 (BAA00012)
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as leisure activities, important life events, emotion, attitude, physical appearance description, travel plans, dreams, countries, people, and languages. Learners can understand and use new languages in a natural, communicative way.

	Learners will be able to present their opinions about some social and cultural issues and understand dialogues and talks. Learners will be able to comprehend 500-700 words passages of familiar topics and gain more knowledge of different cultures around the world. Learners can write paragraphs about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English majors at the University of Science - Vietnam National University - Ho Chi Minh City, using the first six modules in the book New Cutting Edge (Intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics, namely leisure activities, important life events, feelings and emotions, attitudes, physical appearance descriptions, travel plans, presenting dreams, countries, people, and languages. Students need to complete various tasks, including presentations, debates, role-plays, doing homework, tests and so on.
Examination forms	Writing (Midterm: 30%, Final exam: 70%)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, pre-intermediate: workbook*. Harlow: Pearson Education. 3. Materials prepared by the lecturer.

English 4 - BAA00014

Module designation	English 4
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Lecturers at Foreign Language Centre of University of Science, VNU-HCM
Language	English
Relation to curriculum	Compulsory

Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 120 hours. Contact hours: lectures 30 hours; 30 exercise hours. Private study: 60 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • English 3 (BAA00013)
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their intermediate knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners can understand and use the language needed in more complex real-life situations in a natural, communicative way. Learners will be able to express their own ideas in interviews, mini-talks, problem-solving and storytelling. Learners will be able to comprehend 700-1000 words passages of up-to-date topics of international interest and learn more about the world and other cultures. Learners can write essays about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English majors at the University of Science - Vietnam National University - Ho Chi Minh City, using six modules (modules 07-12) in the book New Cutting Edge (Intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics namely everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, and finance-related issues. Students need to complete various tasks, including presentations, debates, role-plays, doing homework, tests and so on.
Examination forms	None
Study and examination requirements	Mid-term test: 50%

	Final test: 50%
Reading list	1. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, Intermediate: student's book*. Harlow: Pearson Education. 2. Sarah Cunningham, Peter Moor, Jane Cornyns Carr (2005). *New Cutting Edge, Intermediate: workbook.* Harlow: Pearson Education. 3. Materials prepared by the lecturer (2012). *Collins Skills for the TOEIC test: Speaking and Writing*. Harper Collins UK.

Environmental Studies - ENV00001

Module designation	Environmental Studies
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at Faculty of Environment, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	90 hours Contact hours: lectures 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This is a compulsory subject in the general knowledge block in the training program for students of all disciplines. The focus of this module is to provide basic knowledge of Environmental Science: concepts, classification of resources, environment, basic problems and core of the environment. Students are also introduced to measures to protect the environment, conserve resources towards the goal of sustainable development.
Content	Chapter 1: Overview of the Environment

	General concepts of environment Chapter 2: Basic composition of the environment (volumes) Chapter 3: Natural Resources 3.1 Definition 3.2 Classification 3.3 General issues of natural disaster Chapter 4: Human Impact on the environment 4.1 History of human impact on the environment 4.2 Human impact on environmental components Chapter 5: Environmental issues and sustainable development 5.1 Population and environment 5.2 Environmental pollution 5.3 Climate change 5.4 Sustainable development Chapter 6: Environmental management and Environmental Education
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm test: 30% Final test: 70%
Reading list	Textbooks: 1. *Lecture on General Environment*. (in Vietnamese). compiled by a group of teachers. 2. Le Van Khoa (Editor), 2004. *Environmental Science* (in Vietnamese). Education Publishing House. References 1. Le Van Khoa, Doan Van Canh, Nguyen Quang Hung, Lam Minh Triet (2011). *Textbook of People and the Environment*. (in Vietnamese). Education Publishing House. 2. Goudie, A. (2006) *The Human Impact on Natural Environment*. 6th Edition. Oxford, Blackwell. 3. Le Thi Thanh Mai (2008), *Textbook of People and the Environment*. VNU-HCM Press.

General Biology I - BIO00001

Module designation	General Biology I
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Biology and Biotechnology, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	This course introduces the topic, gives questions, asks students to see the clip, discusses and reviews
Workload (incl. contact hours, self-study hours)	Total workload: 135 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 45 - Practical hours: 0 - Exercise hours: 0 - Extra-contact hours per week: 0 Private study including examination preparation, specified in hours: 90 hours
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Know the fundamental concepts of biology and the scientific process; distinguish observation, hypothesis, test, and theory Explain the chemical composition of living matter, and chemical reaction related to biology

	Describe the structure and properties of water, structure and function of large biology molecules Distinguish the structure and function of cells, membranes and cellular organelles Describe metabolism at the cellular level, including enzyme activity, cellular respiration, and photosynthesis. Describe DNA structure and function, including replication and repair, transcription, and translation. Describe the mechanism of genetic: cell division, fertilization, mutation, and genetic variation. Explain Biotechnology Explain the evolution and biodiversity Describe the concepts and mechanisms of the origin of species
Content	Introduction to Biology: The fundamental concepts in biology The scientific methods to study the life Part 1. The chemical of life Part 2. The Cell Part 3. Genetics Part 4. Mechanisms of Evolution Conclusion
Examination forms	Multiple choice test
Study and examination requirements	Students create groups: listen, write, and give questions and answers. 6 Quizzes - Midterm examination - Final examination
Reading list	Textbooks 1. Bui Trang Viet (2012). *Cell Biology* (in Vietnamese). VNU-HCM Press.

	2. Pham Thanh Ho (2011). *Biology: Cell, genetics and evolution* (in Vietnamese). VNU-HCM Press. Supplementary reading 1. Reece J. B., Urry, L. A., Cain M. L. 1., Wasserman S. A., Minorsky P. V., Jackson R.B., & Campbell N. A. (2014). *Campbell Biology* (1st edition), Pearson. 2. Lecture slides and related clips.

General Biology II - BIO00002

Module designation	General Biology II
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Biology and Biotechnology, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecturing, posing questions for discussions, presenting videos, giving homework
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 135 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 45 - Practical hours: 0 - Exercise hours: 0 - Extra-contact hours per week: 0 Private study including examination preparation, specified in hours: 90 hours
Credit points	3 Credits (4.5 ECTS)

Required and recommended prerequisites for joining the module	Required prerequisites: None Recommended prerequisites: General Biology I (BIO00001)
Module objectives/intended learning outcomes	PART 1. PLANT MORPHOLOGY AND FUNCTION Learners get to know about plant morphology, functions, and their ecological relationships. They explore nutrient absorption, material transport in vascular plants, and the vital role of soils in providing nutrients. Additionally, they examine plant life cycles, reproductive systems, and how plants respond to environmental cues for survival. PART 2. ANIMAL MORPHOLOGY AND FUNCTION Learners explore the correlation between animal anatomy and physiology. They delve into the structure and functions of key systems including digestion, circulation, respiration, immunity, endocrinology, reproduction, and the nervous system. Additionally, they study osmoregulation, excretion processes, and the mechanisms underlying animal sensation and movement. PART 3. ECOLOGY Learners explore ecological factors and their impact on organisms, studying adaptation strategies. They learn about ecosystem components, structural dynamics, functional processes, material and energy cycling, and equilibrium within ecosystems. Concepts of population dynamics, community interactions, and biosphere principles are also covered, along with the significance of biodiversity conservation for sustaining ecosystems locally in Vietnam and globally.
Content	Part 1. Plant Morphology and Function Part 2. Animal Form and Function Part 3. Ecology
Examination forms	Multiple choice test
Study and examination requirements	None

Reading list	1. Begon M., J.L. Harper, and C.R. Townsend (1986). *Ecology: Individuals, Populations and Communities*. Blackwell Scientific Publications. 2. Brooker, R.J, Widmaier E.P; Graham L.E and Stiling P.D. (214). *Biology*. McGraw-Hill. 3. Campbell N.A, J.B. Reece, L.A. Urry, M.I. Cain, S.A. Wasserman, P.V. Minorsky, and R. B. Jackson (214). *Biology* (1st Edition). Pearson, Benjamin Cummings. 4. Stiling P. D. (22). *Ecology: Theories and Applications* (Fourth Edition). Prentice-Hall, Inc. 5. Smith A.M; Coupland G; Dolan L; Harberd N; Jones J; Martin C; Sablowski R; and Amey A. (21). *Plant Biology*. Garland Science.

General Chemistry 1 - CHE00001

Module designation	General Chemistry 1
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours (in class) Exercise: 30 hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This subject is the first Chemistry subject for students of Chemistry and Materials Science. The subject deals with the theoretical foundations of Chemistry related to the basic models of the atomic structure, the periodic changes in the properties of chemical elements, the fundamental forces of

	interaction in the matter, and the influence of chemical elements. their influence on the properties of matter in the solid, liquid, and gaseous states.
Content	1. Describe the structure of atoms and molecules. 2. Explain periodic changes in some properties of chemical elements. 3. Identify and distinguish basic types of chemical bonds. 4. Identify and explain the relationship between the fundamental forces of interaction in matter and the physical properties of matter.
Examination forms	Written exam Midterm test: 30% Final test: 70%
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	Textbooks: 1. Nguyen Dinh Chi (2007). *General Chemistry* (in Vietnamese). Hanoi Education Publishing House. 2. Nguyen Dinh Soa (2000). *General Chemistry* (in Vietnamese). VNU-HCM Press. 3. Petrucci, R.H; Harwood, W.S; Herring, F.G (2002). *General Chemistry*. 8th Ed. USA: Prentice Hall. Others: 1. Le Thi So Nhu. Summary of General Chemistry lecture - internal documents (For internal circulation only).

General Chemistry 2 - CHE00002

Module designation	General Chemistry 2
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese

Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours (in class) Exercise: 30 hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduce students to the facts, concepts, and vocabulary of chemistry that will enable them to make informed decisions on issues that involve chemical concepts. Give opportunity to use the scientific method to formulate explanations for events in the natural world. Explore the role of measurement in the development of these explanations and the use of logic to test the validity of claims. Show relationships of chemistry to other disciplines, such as mathematics and the biological sciences. Demonstrate the application of chemistry concepts in the everyday world. Give opportunity to apply learned concepts in new situations of both practical and theoretical significance. Give opportunity to read selected scientific materials and apply appropriate language to describe and analyze scientific events. Develop students' scientific problem-solving skills through sequential, mathematically oriented assignments.
Content	The contents are chemical thermodynamics, chemical kinetics, general chemical equilibria, acid and base equilibria, pH, buffers, ionic compound solubility and equilibria electrochemistry.
Examination forms	Written exam Midterm test: 35% Quiz or Assignment: 15% Final test: 50%

Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	1. Ralph H. Petrucci, F. G. Herring, J. Madura, C. Bissonnette (2011), *General Chemistry, Principles and Modern Applications*. 10th Edition. Pearson Canada. 2. Nguyen Dinh Soa (2000). General Chemistry (in Vietnamese). VNU-HCM Press. 3. Petrucci, R.H; Harwood, W.S; Herring, F.G (2002). General Chemistry. 8th Ed. USA: Prentice Hall.

General Chemistry Laboratory 1 - CHE00081

Module designation	General Chemistry Laboratory 1
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Exercise: 60 hours Private study: 30 hours (self-study)
Credit points	2 Credits (4 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Chemistry 1 (CHE00001)
Module objectives/intended learning outcomes	1. Describe the chemical processes of experiments 2. Use the equipment in chemistry lab correctly 3. Calculate, write chemistry laboratory report correctly 4. Work in the lab safely, scientifically
Content	The objective of the general chemistry laboratory course is to become proficient in techniques used by practicing chemist, to use and handle safely and properly laboratory glassware, to carry out experiments safely and carefully in the laboratory, to obtain data accurately and to manipulate

	the data correctly. This course also complements and consolidates the theoretical knowledge acquired in the general chemistry lecture course. In as much as this course is only a supplement to general chemistry lecture courses, students must have had the general chemistry lecture.
Examination forms	Written exam Midterm test: 20% Experimental Reports: 60% Final test: 20%
Study and examination requirements	Minimum lab hours: 7/9 If score of the final exam ≤ 3, the final score will be (midterm test) *20% + (Average of Experimental Reports) *30% + (Final exam) *20%.
Reading list	1. General Chemistry Laboratory 1, Hoàng Ngọc Cường (2023). 2. General Chemistry Laboratory 1-Prelab-Report, Hoàng Ngọc Cường (2023).

General Chemistry Laboratory 2 - CHE00082

Module designation	General Chemistry Laboratory 2
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Lecturers at Faculty of Chemistry, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Exercise: 60 hours Private study: 30 hours (self-study)
Credit points	2 Credits (4 ECTS)

Required and recommended prerequisites for joining the module	Required prerequisites: • General Chemistry 2 (CHE00002)
Module objectives/intended learning outcomes	1. Describe the chemical processes of experiments 2. Use the equipment in chemistry lab correctly 3. Calculate, write chemistry laboratory report correctly 4. Work in the lab safely, scientifically
Content	General Chemistry Laboratory 2 is an introductory course to the general chemistry laboratory. The overall goal of this course is to introduce general chemistry focused on understanding the concepts within the labs and the scientific method. These concepts include but are not limited to understanding basic operation in a chemistry laboratory, components of matter, stoichiometry, determination of concentration by titration, weight measuring, preparation and properties of inorganic and organic substances.
Examination forms	Written exam Midterm test: 20% + Experimental Reports: 60% + Final test: 20%
Study and examination requirements	Minimum lab hours: 7/9 If score of the final exam ≤ 3, the final score will be (midterm test) *20% + (Average of Experimental Reports) *30% + (Final exam) *20%.
Reading list	1. Hoang Ngoc Cuong, *General Chemistry Laboratory 2 Manual*. (in Vietnamese).

General Law - BAA00004

Module designation	General Law
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, seminars
Workload (incl. contact	Total workload: 135 hours

hours, self-study hours)	Contact hours (lectures): 45 hours Private study including examination preparation, specified in hours: 90 hours
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: General Objective: Understand the basic legal concepts and terms related to the country's legal system and state apparatus; apply legal provisions to solve some simple case studies; help students form and develop some skills such as looking up legal documents, analyzing legal regulations, and working in groups, thereby improving their sense of survival, learning and working following the Constitution and regulations. The law, the right behavior orientation in life. Specific objectives/course output standards: Knowledge: Present basic legal concepts and terms related to the state apparatus and the Vietnamese legal system; Solve some exercise cases based on the provisions of a law book in the legal system of Vietnam. Skills: Analyzing legal regulations; Lookup legal documents; Working group. Attitude, diligence: Raise awareness of living, studying, and working following the Constitution and the law.
Content	The module provides knowledge about the structure of the State apparatus as well as the functions, authority, and legal status of agencies in the State apparatus of the Socialist Republic of Vietnam in terms of economic management, Legal nature, and structure of the system of legal documents. From an overview of the system of legal branches in our State's legal system, a course is devoted to studying the basic contents of administrative law, civil law, and criminal law as branches of law. the main law (original branches of law) of the legal system, so that learners can easily access themselves to other branches of law arising from these major branches of law.

Examination forms	Written exam, Multiple choices, Oral presentation
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	1. *General Law textbook*, Ho Chi Minh City University of Law. 2. *Textbook of Theory of State and Law*, Hanoi University of Law.

General Physics 1 (Mechanics and Thermodynamics) - PHY00001

Module designation	General Physics 1 (Mechanics and Thermodynamics)
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Prof. Chau Van Tao (Faculty of Physics and Engineering Physics, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, discussion.
Workload	135 Hours Contact hours: Lectures: 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course covers the principles of kinematics, dynamics, statics, work, energy, linear momentum, gravitation, and thermodynamics. Students who complete this module could achieve the following: Knowledge: Be able to understand and apply laws of mechanics to explain physical phenomena and solve problems; Be able to understand and apply mechanisms of heat transfer, equations of state, the first and the second law of thermodynamics.

	Skills: Be able to work at an individual level and group work. Competence: Ability to apply mechanics and thermodynamics knowledge to analyze physical situations. Attitude: Honest
Content	This module includes the following topics: 1. Physics and measurement 2. Kinematics of particles 3. Force and Newton's laws 4. Conservation laws in classical mechanics 5. Kinetics of rigid bodies 6. The ideal gas 7. The first law of thermodynamics 8. The second law of thermodynamics
Examination forms	Class discussion. Quizzes and projects. Mid-term and Final exam: Written exam (closed book).
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	Main books: 1. Nguyen Nhat Khanh (2005). *Mechanics and thermodynamics lectures*. VNU-HCM Press. References: 1. Nguyen Thanh Van. (2013). *General Physics 1*. VNU-HCM Press. 2. Raymond A. Serway, John W. Jewett, Sr, (2014). *Physics for Scientists and Engineers with Modern Physics*. Brooks/Cole Publishing Company, USA. 3. Alan Giambattista, Betty McCarthy Richardson, Robert C. Richardson, (2010). *Physics*. McGrawHill Companies, Inc, USA.

General Physics 2 (Electromagnetism - Optics) - PHY00002

Module designation	General Physics 2 (Electromagnetism - Optics)

Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Assoc. Prof. Huynh Truc Phuong (Faculty of Physics and Engineering Physics, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lesson, discussion
Workload	135 Hours Contact hours: Lectures: 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Physics 1 (PHY00001)
Module objectives/intended learning outcomes	This module provides basic knowledge of electric and magnetic fields and thereby an understanding of the laws and phenomena of light optics. Students who complete this module could achieve the following: Knowledge: Be able to understand and apply knowledge of electromagnetism and optics in science and life. Skills: Be able to work at individual level and teamwork. Competence: Ability to apply electromagnetism and optics knowledge to analyze physical situations. Attitude: Honesty and diligence
Content	This module includes the following topics: 1. Electric charge and electric field 2. Conductors in an electric field 3. Electric current and magnetic field 4. Electromagnetic induction and applications 5. The background of light optics 6. Interference of light 7. Diffraction of light 8. Polarization of light
Examination forms	Oral presentation

	Mid-term and Final exam: written exam (closed book)
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	Main books: 1. Nguyen Thanh Van. (2015). *General Physics*. (in Vietnamese). VNU-HCM Press. References: 1. Le Vu Tuan Hung (2015). *Optics*. (in Vietnamese). VNU-HCM Press. 2. Raymond A. Serway, John W. Jewett, Sr (2014). *Physics for Scientists and Engineers with Modern Physics*. 9th Edition. BROOK/COLE, USA. 3. Alan Giambattista, Betty McCarthy Richardson, Robert C. Richardson (2010). *Physics*. 2nd Edition. McGrawHill, USA.

History of Vietnamese Communist Party - BAA00104

Module designation	History of Vietnamese Communist Party
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, lesson, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	In terms of content: providing systematic and basic knowledge about the birth of the Communist Party of Vietnam (1920-1930), the Party's leadership over the Vietnamese revolution during the period of political struggle. government authority (1930-1945), in two resistance wars against French colonialism and American imperialism (1945-1975), in the cause of national construction and defense during the country's transition to socialism. association, conducting renovation work (1975-2018). Ideologically: Through historical events and experiences on the leadership of the Party to build a sense of respect for objective truths, raise pride and confidence in the Party's leadership cause. About skills: Equipping with scientific thinking methods on history, skills in choosing research materials, studying subjects and the ability to apply historical awareness to practical work, criticizing misconceptions on the history of the Party.
Content	The course provides systematic and fundamental knowledge about the birth of the Communist Party of Vietnam (19201930), the Party's leadership for the Vietnamese revolution during the period of struggle for power (1930-1945), during the two resistance wars against the French colonialists and the American imperialists (1945-1975), in the cause of national construction and defense during the period of the whole country's transition to socialism, conducted doi moi (1975-2018). Through historical events and experiences in the leadership of the Party to build a sense of respect for objective truth, heighten pride and confidence in the Party's leadership cause. Equip with scientific thinking methods on history, skills in choosing research materials, studying subjects, and the ability to apply historical awareness to practical work, and criticize misconceptions about the history of the Party.
Examination forms	Class discussion. Group presentations and reports.

	Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations for group presentations Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam. Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have test scores, lively discussions, constructive and serious statements in class.
Reading list	*Curriculum of the History of the Communist Party of Vietnam*, Issued by the Ministry of Education and Training.

Ho Chi Minh's Ideology - BAA00003

Module designation	Ho Chi Minh's Ideology
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, lesson, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours:

	Lectures:30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	About knowledge: Equip students with basic knowledge about the concept, origin, the process of formation and development of Ho Chi Minh thought; the basic contents of Ho Chi Minh's thought; the application of the Communist Party of Vietnam in the national-democratic revolution and the socialist revolution, in the current national renewal process. About skills: Helping students to think, analyze, evaluate, and creatively apply Ho Chi Minh's Thoughts to solve problems in real life, study, and work. About attitudes: Helping students improve their political bravery, patriotism, loyalty to the goal, the ideal of national independence associated with socialism; aware of the role and value of Ho Chi Minh's thought for the Vietnamese Party and nation; realize their responsibility in studying and training to contribute to the construction and defense of the country.
Content	Description of course content: the subject equips students with basic knowledge about objects, research methods, and learning meanings of Ho Chi Minh's ideology; on the basis, of the process of formation and development of Ho Chi Minh thought; on national independence and socialism; on the Communist Party and the State of Vietnam; on great national and international solidarity; about culture, ethics, people.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: Multiple choice (closed-book) or essay (opened-book). Final exam: Essay (opened book).
Study and examination requirements	Regulations for group presentations

	Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam. Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance, and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have test scores, lively discussions, constructive and serious statements in class.
Reading list	1. *Textbook of Ho Chi Minh's Thoughts*, National Political Publishing House of Vietnam. 2. *Study Guide for Ho Chi Minh's Thoughts*. VNU-HCM Press.

Human and Environment - ENV00003

Module designation	Human and Environment
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Le Cong Man (Faculty of Biology and Biotechnology, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, discussion
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 30 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 30 - Practical hours: 0 - Exercise hours: 0

	- Extra-contact hours per week: Private study including examination preparation, specified in hours: hours
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: General Biology Recommended prerequisites: 0
Module objectives/intended learning outcomes	Providing students with knowledge about humans' relationship with nature, including inanimate factors and other living things around humans Knowledge: knowing the position, tasks, and roles of humans in the ecosystem. Skills: recognizing results from human activities and impacts on natural and social life. Attitude, Diligence Able to practice soft skills (teamwork, presentations)
Content	Part 1: The formation of human community and society Chapter 1: Introduction Chapter 2: Ecological factors Chapter 3: Population - biome - ecosystem Chapter 4: People and the environment Chapter 5: History of human impact on the environment Part 2: Needs and products in human society Chapter 6: Essential human needs Chapter 7: Resources Chapter 8: Pollution Chapter 9: Environmental protection and sustainable development Chapter 10: Cultural and legal issues.

	Part 3: Observe and identify (from an ecological point of view) ongoing phenomena affecting the lives of humans and other living things.
Examination forms	Mini projects
Study and examination requirements	Study Requirements: 0 Examination Requirements: 1. Discuss 2. Teamwork 3. Seminar 4. Learning attitude 5. Final exam
Reading list	1. Pham Xuan Hau (1997). *Human and Environment*. (in Vietnamese). Vietnam Education Publishing House. 2. Le Van Khoa (2002). *Environmental Science*. (in Vietnamese). Vietnam Education Publishing House. 3. Nguyen Huu Nhan, Hoang Quy Tinh (2011). *Human Ecology Textbook*. (in Vietnamese). Vietnam Education Publishing House.

Introduction to Informatics - CSC00003

Module designation	Introduction to Informatics
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at Centre of Informatics, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, problem, lab works
Workload (incl. contact hours, self-study hours)	Total workload: 165 hours Contact hours: - Lecture hours: 15 - Practical hours: 60 - Exercise hours: 0

	Private study: 90 hours (self-study)
Credit points	3 Credits (5.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: 1. Explain common concepts and terms related to the field of computer science. 2. Perform basic operations related to the Windows operating system and common software. 3. Create documents with professional format, structure, and presentation using software. 4. Analyze and organize data in spreadsheet format, use calculation, search, and statistical functions to create formulas for data processing and presentation. 5. Create eye-catching presentations. 6. Express concepts related to the Internet, information security on the network. 7. Be aware of searching for information and legal regulations regarding copying, sharing, and posting information on the Internet. 8. Design a basic electronic information page.
Content	• Basic understanding of IT • Basic knowledge of computers and computer networks. • Access control, ensuring data security, Malware. • Basic legal issues in using IT. • Basic computer usage • Windows operating system • Windows Explorer • Control Panel • Data compression and extraction • Typing in Vietnamese • Basic Microsoft Word • Document composition • Text formatting • Creating tables • Handling graphics in documents • Page layout and printing • Basic Microsoft PowerPoint • Basic presentation templates • Creating a presentation • Setting up effects for the presentation

	• Basic Microsoft Excel • Data formatting in Excel • References in Excel • Basic Excel functions • Printing and creating charts • Internet usage • Basic knowledge of the Internet • Information searching • Information security • Web image processing • Resizing image frames • Image cropping • Image rotation and flipping • Adjusting the brightness of an image • Adjusting the contrast of an image • Designing a basic electronic information page using HTML & CSS3.
Examination forms	Multiple choice exam, practical exam
Study and examination requirements	Minimum attendance at lectures is 80% Final score is greater or equal to 5.0/10.0
Reading list	1. *Curriculum for Basic IT Applications*, Advanced IT Applications. 2. *Microsoft Office MOS materials*, IIG Vietnam, Fahasha. 3. *IC3 Spark materials*, IIG Vietnam, Fahasha.

Labwork on General Biology I - BIO00081

Module designation	Labwork on General Biology I
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Luong Thi My Ngan (Faculty of Biology and Biotechnology, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture Introducing laboratory rules Divide students into groups of three or two

	Guide experimental procedure Monitor and examine students doing experiments
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 30 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 0 - Practical hours: 30 - Exercise hours: 0 - Extra-contact hours per week: Private study including examination preparation, specified in hours:
Credit points	1 Credits (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Biology I (BIO00001)
Module objectives/intended learning outcomes	Prepare temporary slides to observe cell structures and living activities. Use a microscope proficiently. Identify organic compounds (carbohydrates, proteins, lipids, enzymes like amylase). Demonstrate and measure plant respiration. Classify plants (lichens, algae, bryophytes, pteridophytes, gymnosperms, monocots, dicots). Classify animals into major orders. Use basic lab tools (test tubes, beakers, pipettes, etc.) effectively. Collaborate and assist team members to complete tasks on time. Follow lab safety rules and ensure personal and team safety.
Content	Introduction of Syllabus How to use a light microscope

	Cell structure; Osmosis; Chemical Composition of Cells; Enzyme Activity; Respiration; Mitosis; Microbial Diversity. Plant Diversity; Animal Diversity
Examination forms	Lab Activity and Lab Report Lab exam
Study and examination requirements	Study Requirements: Listen Do experiments Making lab report Put tools back in right places Clean up. Examination Requirements: Experimental process & Final exam
Reading list	Main textbook: 1. Lab manual: Bui Trang Viet & Le Thi My Phuoc (Editors) (2016), *Lab work on General Biology I.* (Internal circulation) 2. Class lecture (ppt file) for *Lab work on General Biology 1.* (in Vietnamese). Supplementary: Dickey Jean (2003). *Laboratory Investigations for Biology*. Pearson.

Labwork on General Biology II - BIO00082

Module designation	Labwork on General Biology II
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	MSc. Phan Ngo Hoang (Faculty of Biology and Biotechnology, University of Science, VNU-HCM)
Language	Vietnamese

Relation to curriculum	Elective
Teaching methods	Preaching, guiding, discussion
Workload (incl. contact hours, self-study hours)	(Estimated) Total workload: 30 Contact hours (please specify whether lecture, exercise, laboratory session, etc.): - Lecture hours: 0 - Practical hours: 30 - Exercise hours: 0 - Extra-contact hours per week: Private study including examination preparation, specified in hours:
Credit points	1 Credit (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • General Biology II (BIO00002)
Module objectives/intended learning outcomes	1. Distinguish different plant tissues and organs. 2. Understand and classify plant pigments. 3. Demonstrate environmental impacts on photosynthesis. 4. Explore respiration and fermentation in plants. 5. Identify different animal tissues and prove bioelectricity. 6. Explain hormonal effects on blood flow. 7. Understand climatic impacts on ecosystems. 8. Describe biological reactions and processes. 9. Interpret respiration and photosynthesis. 10. Explain cellular signal transduction. 11. Distinguish plant and animal structures and evolution. 12. Conduct and analyze biological experiments and biodiversity research.
Content	1. Practice theory 2. Tissue and plant organs 3. Plant pigments 4. Photosynthesis 5. Respiration and fermentation in plants

	6. Water transport of tissues 7. Animal tissues 8. Mechanical and bioelectric activities 9. Circulation of blood in the vein 10. Climate factor
Examination forms	Abstract paper: objectives, principles and methods of practice Practice report Practice and question
Study and examination requirements	Listen, take notes, practice, discuss Students must not be absent in practice theory class (the first week) Students wear blouses when entering the Lab Submit practice reports at the end of the practice session Students who are absent for more than 2 practice sessions will be banned from the final exam. Examination Requirements: Homework Lab work Final examination
Reading list	Textbooks: 1. Bui Trang Viet, Phan Ngo Hoang (2009). *Practical Textbook for General Biology 2* (in Vietnamese). VNU-HCM Press. Supplementary reading: 1. Lecture notes. 2. Pham Thanh Ho (1998). *General Biology.* (in Vietnamese). Vietnam Education Publishing House.

General Physics Lab - PHY00081

Module designation	General Physics Lab

Semester(s) in which the module is taught	3rd semester
Person responsible for the module	MSc. Huynh Thanh Nhan (Faculty of Physics and Engineering Physics, University of Science, VNU-HCM)
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lab work, discussion, practice
Workload (incl. contact hours, self-study hours)	120 Hours Contact hours: Lab work: 60 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (4 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course is a practical subject in the laboratory. This course helps students understand how to measure some physical quantities, experimental errors, analyze and evaluate measurement results. Students who complete this module could achieve the following: Knowledge: Be able to describe the process, how to measure fundamental physical quantities in the laboratory. Be able to use instruments and equipment to measure experimental data of physical quantities correctly. Be able to determine (calculate) physical quantities from measured experimental data. Be able to determine the error of experimental measurement of physical quantities. Skills: Be able to work in individual, group work, self-study, and problem solving. Competence: Be able to analyze, process and write experimental data reports. Attitude: be honest, responsible, respect for colleagues.
Content	In this module, Students practice 10 of the following 13 experiments: Practice 1: Density of liquid and solids. The private mass of the metals.

	Practice 2: Viscosity. Viscosity is dependence of different temperatures. Practice 3: Reversible pendulum. The Mathematical pendulum. Practice 4: Heat of function for ice. Determination of heat. Practice 5: Mechanical equivalent of heat. The heat capacity of metals. Practice 6: Wheatstone Bridge. Resistor is dependent of different temperatures. Practice 7: Voltmeter and Ammeter DC. Voltmeter and Ammeter AC. Practice 8: AC circuit. RLC circuit. Practice 9: Diode characteristics. Practice 10: Transistor characteristics. Practice 11: Microscope. To measure the diameter of other small objects. Practice 12: Refraction by a prism. Dispersion and resolving the power of prisms. Practice 13: Polarization of light Rotatory power.
Examination forms	Practice reports, practice exam
Study and examination requirements	Minimum attendance at Lab is 80% (Absences must not exceed 3 times for the entire duration of the lectures) Final score is greater or equal to 5.0/10.0
Reading list	1. Dang Van Liet, Do Dinh Luyen, Nguyen Van Nghia, Tran Thi Kim Phuong (2008). *General Physics Experiments*. (in Vietnamese). VNU-HCM Press.

Marxist-Leninist Philosophy - BAA00101

Module designation	Marxist-Leninist Philosophy
Semester(s) in which the module is taught	1st semester

Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	135 Hours Contact hours: Lectures: 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The course equips students with the basic contents of the worldview and the Marxist-Leninist philosophical methodology. Helping students apply knowledge about the worldview, Marxist-Leninist philosophy, and philosophy creatively in cognitive and practical activities, to solve problems that the social life of a country or of the time being set.
Content	Marxist-Leninist philosophy is a course in the Marxist-Leninist knowledge block and Ho Chi Minh Thought. This module equips students with basic, general, and systematic knowledge of the problems of Marxist-Leninist philosophy. From there, learners have a basis and reasonable research and learning methods and apply them to the process of evaluating life phenomena.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book)
Study and examination requirements	Regulations for group presentations

	Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam. Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have test scores, lively discussions, constructive and serious statements in class.
Reading list	1. *Textbook of basic principles of Marxism-Leninism*, National Political Publishing House of Vietnam. 2. *Textbook of Marxist-Leninist Philosophy*, National Political Publishing House of Vietnam.

Marxist-Leninist Political Economy - BAA00102

Module designation	Marxist-Leninist Political Economy
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Lecturers at School of Political and Administration Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Firstly, equip students with basic and core knowledge of the Marxist-Leninist political economy in the context of economic development of the country and the world today. Ensure the basic, systematic, scientific, updating new knowledge, associate with practice, creativity, skills, thinking, learner quality, connectivity to overcome duplication, enhance integration and reduce the load, reduce content that is no longer relevant or scholastic content for students at non-theoretical colleges and universities. Second, on that basis, forming thinking and analytical skills, assessing and identifying the nature of economic benefit relations in the country's socio-economic development, contributing to helping students build appropriate social responsibility in the job position and life after graduation. Third, contribute to building the stance and ideology of Marxism-Leninism towards students.
Content	The program consists of six chapters: in which chapter one discusses the objects, research methods, and functions of the Marxist-Leninist Political Economy. Chapters 2 to 6 present the core content of the Marxist-Leninist political economy according to the subject's objectives. Specifically, issues such as goods, markets and the role of actors in the market economy; Producing surplus value in a market economy; Competition and monopoly in the market economy; Socialist-oriented market economy and economic interest relations in Vietnam; Industrialization, modernization, and international economic integration in Vietnam.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations for group presentations: Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam.

	Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Regulations on time, attendance, and discipline in the course: attend class on time and at least 80% of the sessions (only to be absent for a maximum of 20%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have all test scores, lively discussions, constructive and serious statements in class.
Reading list	Marxist-Leninist political economy textbook for undergraduates who are not majoring in political economy.

Psychology - BAA00006

Module designation	Psychology
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Tran Huong Thao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total workload: 90 hours. Contact hours (lecture): 30 hours. Private study including examination preparation, specified in hours: 60 hours.
Credit points	2 credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Knowledge:

	• Understand the system of basic concepts of psychological science and research methods in psychology. • Understand the origin, formation and development of psychology and consciousness. • Understand the nature of human psychological processes: perception; emotion - affection; act. • Identify human psychological states. • Understand the psychological attributes that make up the personality structure. Understand the factors affecting the formation and development of personality. Skills: • Developing the capacity to study documents: Analyze, synthesize, compare, and generalize. • Form and develop the ability to identify psychological phenomena and apply learned knowledge to solve practical problems. • Consulting and consulting skills. Attitude: • Cultivate a passion for learning and studying subjects. • Forming a sense of initiative and positivity in self-study. • Form the right motivation in learning. • Raise a sense of responsibility for group activities. • Other goals: • Forming personality qualities in accordance with the requirements of the integration period. • Forming communication and behavioral skills in the community. • Forming a modern and scientific way of living and working. • Forming and developing the ability to think creatively, independently, and critically. • Reasoning skills, public speaking skills. • Form and develop teamwork skills.
Content	The course introduces learners to acquire basic knowledge about the nature and characteristics of psychological phenomena and basic psychological laws of humans (perception, emotion, will, etc.) actions and personalities...). On that basis, it helps learners to apply knowledge in

	practice to identify and distinguish basic psychological phenomena in humans. This module includes the following parts: Part 1: The Natural Basis and Social Foundation of Human Psychology. Part 2: Perception and Learning. Part 3: Emotions - Affections. Part 4: Personality and the Formation of Personality. Part 5: Deviations in Individual Psychological Behavior and Correcting Such Deviant Behavior.
Examination forms	Class discussion; Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations on time, attendance, and discipline in the course: attend class on time and at least 70% of the sessions (only to be absent for a maximum of 30%). Exam ban is applied to those who miss more than the regulated number of sessions. Students must have all test scores, lively discussions, constructive and serious statements in class.
Reading list	Main textbook: 1. Nguyen Quang Uan (2015). *General Psychology: A Textbook*. (in Vietnamese). References: 1. Plotnik, R, Kouyoumdjian, H (2011). *Introduction to Psychology*. 2. Berstein, D. A., Penner, L. A., Clarke-Stewart, A., and Roy, E (2008). *Psychology*.

Scientific Socialism - BAA00103

Module designation	Scientific Socialism
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Lecturers at School of Political and Administration

	Sciences, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	lecture, lesson, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ETCS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The subject equips students with the basic contents of scientific socialism (one of the three components constituting Marxism-Leninism). Helping students apply basic knowledge of scientific socialism creatively in cognitive and practical activities, solving problems that the social life of a country, of the times being set.
Content	The subject equips students with the basic contents of scientific socialism (one of the three components constituting Marxism-Leninism). Helping students apply basic knowledge of scientific socialism creatively in cognitive and practical activities, solving problems that the social life of a country, of the times being set.
Examination forms	Class discussion. Group presentations and reports. Mid-term exam: essay (opened book). Final exam: essay (closed book).
Study and examination requirements	Regulations for group presentations Forming a group: 5 students/group. The deadline for group topic registration on the forum is session 2 or directly submit it to the lecturer at the exam.

	Week 4 (4th session) begins to present in order. Note that the presenting groups need to fully show up and bring along all relevant documents. Submission form: submit files and minutes of group work via email to the lecturer. Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures)
Reading list	1. *Textbook of Scientific Socialism*, National Political Publishing House of Vietnam. 2. *The Basic Principles of Marxism-Leninism*, National Political Publishing House of Vietnam.

Teamwork and Learning Skills - BAA00008

Module designation	Teamwork and Learning Skills
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Tran Huong Thao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Do lecturing, teamwork, divide students into groups to solve problems
Workload (incl. contact hours, self-study hours)	Total: 60 hours Contact hours: Lectures 30 hours (in-class). Private study: 30 hours (self-study).
Credit points	2 credits (3 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: Specific objectives/subject output standards: Equip students with knowledge and skills about teamwork and

	study skills. Bringing career orientations to group work practice topics. Skills: teamwork, presentation, communication Attitude, diligence: serious, diligent, positive
Content	Understand the concept of groups, how to form groups, classify groups, functions, and tasks of group members. Knowing the process of teamwork, the necessary skills when working in groups. Able to plan and execute projects. Understand and apply learning skills.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Team Leadership Techniques*. Tre Publishing House. 2. *Strategic Learning*. Labour & Social Publishing House.

Analysis 1A - MTH00010

Module designation	Analysis 1A
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Assoc. Prof. Dr. Ly Kim Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: lectures: 30 hours (in class); 30 exercise hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	The objective of the module is to equip students with the basic knowledge of the foundation of calculus as the foundation for specialized modules.
Content	The course covers the basics of real numbers, sequences and series of real numbers, one variable continuous function.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Mathematica by example*, Academic Press, New York 2. *Calculus*, Harcourt Brace College Publishers, New York 3. *Textbook of Calculus I*. Statistical Publishing House, Ho Chi Minh City. 4. *Analysis 1*, Tenence Tao, 2022, Sringer.

Analysis 2A - MTH00012

Module designation	Analysis 2A
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Le Anh Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours *(in class).* Exercises: 30 hours *(in class)* Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites • Analysis 1A (MTH00010)

Module objectives/intended learning outcomes	General objective: To introduce the basic concepts and techniques of metric topology for multivariable functions and vectors. Specific objectives: Knowledge: know the basic concepts of metrics, closed, open, compact sets, completeness, series in normed space. Skills: do exercises that demonstrate the above concepts and their applications
Content	This module helps students understand basic topological concepts, understand convergence in multi-dimensional spaces and function spaces. The course introduces metric spaces, normative spaces and Rn; properties of continuous functions on metric spaces. Completeness of spaces and series on complete normed space. This knowledge forms the basis for all specializations such as analysis, statistics and probability, computer science, optimization, and applied mathematics.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 20%, Midterm Exam: 30%, Final Exam: 50%
Reading list	1. Dang, D. T., Dinh, N. T., Pham, H. Q. (2005). *Calculus A2* (in Vietnamese). VNU-HCM Press. 2. Tenence Tao. (2022). *Analysis 2*. Springer. 3. Nancy, G. P., Scott, F. (1985). *Calculus*. Harcourt Brace College Publishers, New York.

Analysis 3A - MTH00014

Module designation	Analysis 3A
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Huynh Quang Vu
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises

Workload (incl. contact hours, self-study hours)	165 Hours Contact hours: lectures 45 hours + 30 exercises hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis 1A (MTH00010) • Calculus 1A (MTH00011) • Analysis 2A (MTH00012)
Module objectives/intended learning outcomes	General objective: Study the integral of functions of many variables and the relationships between the differential and integral of functions of many variables. Specific objectives: Knowledge: The minimum outcome standard is at the level in J. Stewart's Calculus textbook for science and engineering students. The average level is more advanced, more suitable for Mathematics majors, with higher requirements for accuracy and theoretical content. For fairly good students and honor students the course aims for qualifications in the respective sections of classic analysis textbooks such as those of W. Rudin, S. Lang. Skills: Introduction to computer tools. Exercises include both reasoning and calculation. Attitude, diligence: Seeing the need to develop generalization and precision, forming an ability to solve new application problems. Having a serious, proactive and self-disciplined learning attitude.
Content	This is a course on Multiple Integration and Vector Calculus. This subject follows the subjects of Calculus 1 and Calculus 2, which is considered a basic knowledge for university level in Science and Technology. A useful course for more advanced investigations of Lebesgue integrals (Measurement and Probability), mathematical models using Integral (in Mechanics, Probability-Statistics, Mathematical Equations, Calculus, ...), and mathematical developments (in Analysis, Geometry, ...)

Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 20%, Midterm Exam: 30%, Final Exam: 50%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Calculus III Textbook*, Huynh Quang Vu, VNU-HCM Press, 2017. Class notes available at https://sites.google.com/view/hqvu/teaching 2. *Calculus: Early Transcendentals*. James Stewart, 8th ed., Brooks/Cole, 2016. 3. *Advanced Calculus*. Gerald B. Folland, Pearson, 2001.

Ordinary Differential Equations - MTH00015

Module designation	Ordinary Differential Equations
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Dr. Le Anh Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	135 hours Contact hours: lectures 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis 1A (MTH00010) • Calculus 1A (MTH00011) • Analysis 2A (MTH00012) • Calculus 2A (MTH00013)
Module objectives/intended learning outcomes	General objective: To introduce basic differential equations and know how to solve basic ordinary differential equations. Specific objectives/course outcome standards:

	Knowledge: Master the course content. Skills: Ability to calculate with large numbers of calculations Attitude, diligence: Diligent and serious
Content	This module equips math majors with a minimum of background knowledge before entering narrower specializations. The content of this course includes: Differential equations of first order. Existence and uniqueness of solutions of Cauchy problem. Linear differential equations of second and higher order. Introduction to the system of differential equations of first order.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm Exam: 20%, Final Exam: 80%
Reading list	1. Nguyen Thanh Long (2012). *Lecture Notes on Calculus A4* (in Vietnamese). 2. Nguyen, T. L., Nguyen, C. T., Le, T. P. N., and Nguyen, A. T. *Differential Calculus of Multivariable Functions and Differential Equations* (in Vietnamese). 3. Ravi, P. A., Donal, O'R. (2008). *An introduction to ordinary differential equations*. Springer.

Calculus 1A - MTH00011

Module designation	Calculus 1A
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Assoc. Prof. Dr. Ly Kim Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours:

	Lectures: 30 hours (in class); 30 exercise hours Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The objective of the module is to equip students with the basic knowledge of calculus as the foundation for specialized modules.
Content	The course covers the basics of classic differentiation, and Riemann integrations.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Mathematica by example*, Academic Press, New York 2. *Calculus*, Harcourt Brace College Publishers, New York 3. *Textbook of Calculus I*. Statistical Publishing House, Ho Chi Minh City 4. *Analysis 1*, Tenrence Tao, 2022, Springer.

Calculus 2A - MTH00013

Module designation	Calculus 2A
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Le Anh Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: lectures: 30 hours (in class); 30 exercise hours Private study: 90 hours (self-study)

Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 1A (MTH00011)
Module objectives/intended learning outcomes	General Objective: To introduce the basic concepts and techniques of differentiable calculus of multivariable vector functions. Specific objectives: Knowledge: The theory of a series of real numbers. Properties of differentiable functions on Rn. Skills: be able to do calculation exercises on the above concepts and their applications
Content	This module helps students understand the basic knowledge of differential calculus of multivariable functions, understand bound and unconstrained extremal problems in multidimensional spaces, and understand the concept of series of real numbers. This knowledge forms the basis for all specializations such as analysis, statistics and probability, computer science, optimization, applied mathematics.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 20%, Midterm Exam: 30%, Final Exam: 50%
Reading list	1. Dang, D. T., Dinh, N. T., Pham, H. Q. (2005). Calculus A2 (in Vietnamese). VNUHCM Press. 2. Nancy, G. P., Scott, F. (1985). Calculus. Harcourt Brace College Publishers, New York. 3. Sudhir, R. G., Balmohan, V. L. (2010). A Course in Multivariable Calculus and Analysis. Springer.

Computational Software Laboratory - MTH00087

Module designation	Computational Software Laboratory
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Ong Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory

Teaching methods	Lectures, computer practices
Workload (incl. contact hours, self-study hours)	120 hours Contact hours: 60 lab hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (4 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Students can use computational programming languages to program calculations in mechanics, analysis, numerical analysis, algebra, statistics…
Content	The course covers basic computational programming knowledge using Matlab... Applying this knowledge to write programs to solve problems in mechanics, analysis, numerical analysis, algebra, ...
Examination forms	Tests - process score: 30% Midterm exam: 30% Final exam: 40%
Study and examination requirements	Midterm and final exam: practical exercises in the laboratory. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Brian, D. H., Daniel, T. V. (2007). *Essential MATLAB ® for Engineers and Scientists*, 3rd edition, Elsevier. 2. J. H. Mathews, K. D. Frink, (1999). *Numerical methods using Matlab*. 3rd Edition. Prentice Hall. 3. Ronald, L. L., Jonathan, M. R. (2017). *Multivariable Calculus with MATLAB: With Applications to Geometry and Physics*. Springer.

Linear Algebra - MTH00030

Module designation	Linear Algebra
Semester(s) in which the module is taught	1st semester
Person responsible for the module	Dr. Le Van Luyen

	Dr. Bui Anh Tuan
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	135 hours Contact hours: lectures 45 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General objectives: To be familiar with advanced mathematics. Specific objectives/course learning outcomes: Knowledge: Mastering the knowledge of matrices on number fields and applying them to solving systems of linear equations; determinants and their applications; vector spaces and linear transformations. Skills: Calculating matrices; solving system of linear equations; calculating the coordinates of the vector according to a basis in a finite dimensional vector space; changing when changing the base; representing a linear operator by a matrix; finding the image and the kernel of linear operators; using MAPLE software. Attitude: Attend all classroom sessions; discussions outside of class time.
Content	This course is taught in the first semester, initially introducing students to advanced mathematics. Besides equipping new knowledge that is necessary for freshman, this course also provides the foundation knowledge to help students carry out specialized courses.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm: 40% Final: 60%

Reading list	1. Bui, X.H., Tran, N.H., Trinh, T.D, Le, V. L. (2009). *Linear Algebra and Its Applications*, Vol 1 (in Vietnamese). VNUHCM Press. 2. Ngo, V. T. (2001). *Textbook of Linear Algebra*. (in Vietnamese). Vietnam National University Press, Hanoi. 3. David, C. L. (2011). *Linear Algebra and Its Applications*. (in Vietnamese). Pearson Education.

Linear Algebra Practice - MTH00083

Module designation	Linear Algebra Practice
Semester(s) in which the module is taught	1st semester
Person responsible for the module	MSc. Pham The Nhan
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	60 hours Contact hours: exercises 30 hours (in class) Private study: 30 hours (self-study)
Credit points	1 credit (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030)
Module objectives/intended learning outcomes	General objectives: To help students understand more linear algebra via doing exercise. Specific objectives/course learning outcomes: Knowledge: Mastering the knowledge of matrices on numerical fields and applying them to solving systems of linear equations; determinants and their applications; vector spaces and linear transformations. Skills: Calculating matrices; solving system of linear equations; calculating the coordinates of the vector according to a basis in a finite dimensional vector space; changing when changing the base; representing a linear

	operator by a matrix; finding the image and the kernel of linear operators; using MAPLE software. Attitude: Attend all classroom sessions; discussions outside of class time.
Content	This course is taught in the first year, initially introducing students to advanced mathematics. Besides equipping new knowledge that is necessary for freshman, this course also provides the foundation knowledge to help students carry out specialized courses.
Examination forms	Tests - process score: 30% Midterm exam: 30% Final exam: 40%
Study and examination requirements	Midterm and final exam: written exams.
Reading list	1. Bui, X.H., Tran, N.H., Trinh, T.D, Le, V. L. (2009). *Linear Algebra and Its Applications*, Vol 1 (in Vietnamese). VNUHCM Press. 2. Ngo, V. T. (2001). *Textbook of Linear Algebra*. (in Vietnamese). Vietnam National University Press, Hanoi. 3. Pham, H. D. (2009). *Computer, Program and Teach Mathematics on Maple.* (in Vietnamese). Science and Technics Publishing House.

Abstract Algebra - MTH00031

Module designation	Abstract Algebra
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Trinh Thanh Deo
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	135 hours Contact hours: lectures 45 hours (in class) Private study: 90 hours (self-study)

Credit points	3 Credits (4.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030) • Linear Algebra Practice (MTH00083)
Module objectives/intended learning outcomes	The goal of the course is to equip students with basic abstract algebraic structures, to help them become familiar with symbols and formal calculations.
Content	The course introduces basic algebraic structures such as groups, rings, integer domains, fields, and polynomial rings on fields. Some special concepts introduced are permutation group, alternating group, cyclic group, polynomial ring on number fields, especially on rational number fields.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nguyen, V. D., Tran N. H. (2005). *Abstract Algebra* (in Vietnamese). VNUHCM Press. 2. Hoang, X. S. (1997). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House. 3. Nguyen, H. V. H. (1998). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House. 4. My. V. Q. (1998). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House. 5. Bui, H. H., Nguyen H. H., Phan D. T. (1985), *Algebra and Arithmetic Exercises*, Vol 1 and 2. (in Vietnamese). Vietnam Education Publishing House.

Abstract Algebra Practice - MTH00084

Module designation	Abstract Algebra Practice
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	Dr. Nguyen Kim Ngoc
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	lectures, computer practices

Workload (incl. contact hours, self-study hours)	90 hours Contact hours: practice 30 hours (in lab) Private study: 60 hours (self-study)
Credit points	1 Credit (2 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030) • Linear Algebra Practice (MTH00083) • Abstract Algebra (MTH00031)
Module objectives/intended learning outcomes	General Objective: To equip students with basic computational skills in abstract algebraic structures, formal calculation, giving them a better understanding of these algebraic structures. Use software (e.g. GAP) to calculate specific examples and exercises. Specific objectives/course learning outcomes: Knowledge: Master the theory of basic algebraic structures such as groups, rings, fields, and polynomial rings. Get hands-on with software (e.g. GAP) to better understand the structure learned. Skills: Improve reasoning skill and formal calculation. Attitude, diligence: Serious, progressive.
Content	The goal is to practice basic computational skills on algebraic structures such as groups, rings, integral domains, and fields, and use software (e.g. GAP) to better understand the structure of theories learned.
Examination forms	Tests - process score: 30% Midterm exam: 30% Final exam: 40%
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nguyen, V. D., Tran N. H. (2005). *Abstract Algebra* (in Vietnamese). VNUHCM Press. 2. Hoang, X. S. (1997). *Abstract Algebra* (in Vietnamese). Vietnam Education Publishing House.

	3. Alexander Hulpke (2011). *Abstract Algebra in GAP*. Online version.

Introduction to Computer Programming - MTH00055

Module designation	Introduction to Computer Programming
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	MSc. Nguyen Hien Luong MSc. Ha Van Thao
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: lectures 45 hours + 30 lab hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduce to students the logic of computer programming – how computers operate with programming languages and how to write instructions using flowchart or pseudocode; how to modularize a computer program; how to write computer programs in C language.
Content	Basic concepts of algorithms and algorithms. Describe the skills and fundamentals of computer programming. Introduction to the C programming language. Pointer. Array and string. Function. Structure.

Examination forms	Final exam: written exams.
Study and examination requirements	Test 1: 15% Test 2: 15% Test 3: 20% Final exam: 50% Final score is greater or equal to 5.0/10.0
Reading list	Main: 1. Avelino J. Gonzalez, *Computer Programming in C for Beginners*, Springer Cham, 2020. References: 1. G. McKenzie. *Programming Languages: C and C++*. Murphy & Moore Publishing, 2022 2. Orhan Gazi. *Modern C Programming Including Standards C99, C11, C17, C23*. Springer Cham, 2023. 3. Pham Van At. *C and C# Programming Techniques: From Basics to Advanced.* Hanoi: Hanoi University of Science and Technology Publishing House, 2024.

II. PROFESSIONAL EDUCATION KNOWLEDGE

1. BASIC KNOWLEDGE

Algebra 2A - MTH10402

Module designation	Algebra 2A
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Dr. Tran Ngoc Hoi
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Algebra (MTH00030) • Linear Algebra Practice (MTH00083)
Module objectives/intended learning outcomes	General objective: To equip students with basic knowledge of advanced linear algebra, as a foundation for specialized courses in Mathematics. Specific objectives for course learning outcomes: Knowledge: Master the diagonalization and Jordanization of linear operators on finite-dimensional vector spaces, the structure of Euclidean spaces, bilinear forms, and quadratic forms. Skills: Diagonalization and Jordanization of matrices on numeric fields, computation in Euclidean space, quadratic transformations.

	Attitude, diligence: Serious, progressive.
Content	This course introduces the basic knowledge of linear operator reduction (diagonalization, Jordan canonical form), Euclidean space, bilinear form, and quadratic form.
Examination forms	Midterm and Final exam: written exam
Study and examination requirements	Tests - process score: 50% Final exam: 50% Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Bui, X.H., Tran, N.H., Le, V. L. (2017). *Linear Algebra and Its Applications*, Vol 2 (in Vietnamese). VNUHCM Press. 2. Ngo, V. T. (2001). *Textbook of Linear Algebra*. Vietnam National University Press, Hanoi. 3. David, C. L. (2011). *Linear Algebra and Its Applications*. Pearson Education.

Functional Analysis - MTH10403

Module designation	Functional Analysis
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Bui Le Trong Thanh
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, practice sessions
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: lectures 45 hours + 30 exercise hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	General Objective: Functional analysis is where students get their first basic understanding of infinite-dimensional spaces. This knowledge is indispensable for many specializations in both theoretical and applied mathematics. This is one of the first courses where the ability to acquire and use abstract and precise mathematical reasoning is trained and tested. Specific objectives/course outcomes standards: Knowledge: understand and apply in specific situations some basic concepts and results about Euclidean spaces, l^p, L^p spaces, bounded functions spaces, space of continuous linear mappings, Hilbert space. Skills: systematically understanding the above objects, grasping the relationships between concepts and results. Distinguish and criticize arguments that do not meet the exact requirements of mathematics. Attitude, diligence: Seeing the need for generalized development, thereby helping to solve application problems, thereby having a serious, proactive, and self-disciplined learning attitude.
Content	Metric space, normed space, continuous linear mapping between two normed spaces and their fundamental theorems, Hilbert space. This subject is considered as the basic knowledge for university level Mathematics. An essential and useful course for mathematical models using function spaces and infinite dimensional spaces (in Algebra, Optimization, Probability-Statistics, Partial Differential Equations, Analysis, Mathematical Methods in Physics, Computer Science, ...).
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Assignments: 50%; Final Exam: 50% Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Lecture Notes on Functional Analysis*, Dinh Ngoc Thanh and Huynh Quang Vu. 2. *Functional Analysis*, Duong Minh Duc, 2005. 3. *Functional Analysis*, Dinh Ngoc Thanh and Dang Duc Trong, 2011.

Measure Theory and Probability - MTH10401

Module designation	Measure Theory and Probability
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	Prof. Dang Duc Trong
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis A1 (MTH00010) • Calculus A1 (MTH00011)
Module objectives/intended learning outcomes	The objective of the module is to equip students with the basic knowledge of the foundation of measure theory and the theory of integral, and the application of the measure theory in probability theory.
Content	Elementary probability theory, abstract measure theory. Integration with respect to probability measures, expectation, and variance Random variables, Law of large numbers and limit theorems. Changes of measures and the Radon-Nikodym Theorem. Conditional expectations, filtrations, and martingales.
Examination forms	Midterm and final exam: written exams.

Study and examination requirements	Exercises 20%; Mid-term 30%; Final exam 50%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Trong, D. D., Thanh, D. N., Minh, N. D. (2024). *Probability Theory*. (in Vietnamese). VNUHCM Press. 2. Trong, D. D., Thanh, D. N. (2014). *Textbook of Measure Theory and Probability*, (in Vietnamese). VNUHCM Press. 3. K. L. Chung, *A Course in Probability Theory*, 3rd edition, Academic Press, 2001. 4. Quan, P. H., Thanh, D. N. (2011). *Probability and Statistics*. Vietnam Education Publishing House.

Mathematical Statistics - MTH10404

Module designation	Mathematical Statistics
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Dr. Hoang Van Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, exercises, practice sessions
Workload (incl. contact hours, self-study hours)	Total: 145 hours. Contact hours: Lectures: 15 hours (in class) Exercises: 30 hours (in class) Practice: 30 hours (in lab) Private study: 70 hours (self-study)
Credit points	3 Credits (5.5 ECTS)

Required and recommended prerequisites for joining the module	Required prerequisites: • Probability and Statistics (MTH00040) - for students of DS programme. • Measure Theory and Probability (MTH10401) - for students of other programmes.
Module objectives/intended learning outcomes	This course provides foundations of statistical inference. Students will be able to make inferences from data, making decisions and predictions. Students can apply statistical techniques to practical problems using R or SPSS.
Content	Descriptive statistics. Sampling distributions. Point estimations: definition, estimators, and estimates. Unbiasedness, efficiency, consistency. Method of moments, method of maximum likelihood, Bayes estimators. Minimum variance unbiased estimator, Cramer-Rao lower bound. Confidence interval: CI for means, variances and proportions. Hypothesis testing: basic concepts, null and alternative hypotheses, simple and compound hypotheses, type I and II errors, critical region, size and power of a test, p-value. Neyman - Pearson lemma. Likelihood Ratio tests. z-tests and t-tests. Goodness of fit test. Linear Regression.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Homework exercises: 10% Computing exercises: 20% Mid-term: 20%. Final exam: 50%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Casella, George, and Roger L. Berger. *Statistical Inference*. Cengage Learning, 2021.

	2. Lehmann, Erich Leo, Joseph P. Romano, and George Casella. *Testing Statistical Hypotheses*. Vol. 3. New York: Springer, 2005. 3. Montgomery, Douglas C., and George C. Runger. *Applied Statistics and Probability for Engineers*. John Wiley & Sons, 2010.

2. SPECIALIZATION IN MECHANICS

2.1. COMPULSORY COURSES

Numerical Analysis 1 - MTH10410

Module designation	Numerical Analysis 1
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Dr. Ong Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mechanics, AM programme. Elective for others programme.
Teaching methods	Lectures, lab works
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: lectures 45 hours + 30 practical lessons in the laboratory. Private study: 90 hours *(self-study).*
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General Objective: To introduce the basic concepts and approximation methods of numerical analysis, and to provide students with the tools to find approximate solutions to problems involving equations and systems of equations. By finding algorithms to solve the problems posed, this module aims to equip students with knowledge to solve real problems based on theoretical math knowledge and the means of computations. The course helps students understand the following knowledge: the concept of approximation and types of errors. Understand the most basic approximation methods and their

	applications. Application of computational software in numerical calculations.
Content	The content of the course includes the approximate theories and their application to find the approximate solutions of the single variable equations and the system of linear equations, and the approximate derivative, integral... These methods, which are applied to physical problems, are used in the Matlab program to simulate.
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Exam: 30%, Final Exam: 70%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Numerical Analysis*. Ninth Edition. Brooks, Burden and Faires, 2010. 2. *Numerical Analysis*. Pearson, Sauer, 2006. 3. *Numerical Analysis*. Brooks/Cole Publishing Company, Kincaid, W. Cheney, 1991. 4. *Numerical methods using Matlab.* Third Edition. Prentice Hall, J. H. Mathews, K. D. Frink, 1999

Equations of Mathematical Physics - MTH10413

Module designation	Equations of Mathematical Physics
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Le Duc Hung
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures
Workload (incl. contact hours, self-study hours)	Total: 150 hours Contact hours: lectures 60 hours (in-class). Private study: 90 hours (self-study).
Credit points	4 Credits (6.0 ECTS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Introduce the basic mathematical physics equations and know how to find classical solutions by Fourier's method of separation of variables.
Content	Linear ordinary differential equations of the second order. Wave equations; Heat equations; Laplace equations.
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Midterm Exam: 30%, Final Exam: 70% Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Lecture Notes on Mathematical Physics Equations, Nguyen Thanh Long, 2020. 2. Advanced Mathematical Physics Equations, Nguyen Cong Tam, 2002. 3. Partial Differential Equations, An introduction, David Colton, 1988. 4. Partial Differential Equations, L.C. Evans, 1998. 5. Fundamentals of differential equations and boundary value problems, R. Kent Nagle, Edward B. Saff, 1993.

Theoretical Mechanics - MTH10427

Module designation	Theoretical mechanics
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Le Van Chanh
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mechanics, AM programme
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	150 hours

	Contact hours: Lectures: 60 hours (in-class) Private study: 90 hours (self-study)
Credit points	4 credits (6.0 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 2A (MTH00013) • Linear Algebra (MTH00030)
Module objectives/intended learning outcomes	General Objective: To provide basic knowledge of Newtonian mechanics. Help students know how to set and solve basic mechanical problems. Know how to analyze and explain mechanical phenomena based on Newton's laws. The course introduces mathematical modeling of real-world processes. Specific objectives/course learning outcomes: Knowledge: Newton's laws and their consequences. Know how to build mathematical models describing mechanical phenomena. Know how to solve some mechanical problems. Attitude, diligence: Positive and dynamic.
Content	This module includes the following topics: • Vectors • Mechanics of Particles • Mechanics of Particle Systems • Mechanics of Connected Systems
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Tests - process score: 40%; Final exam: 60 % Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Course in Theoretical Mechanics* (in Vietnamese), Le Van Chanh, lecture note, 2022. 2. *Lectures in Classical Mechanics*, V Ilisie, Springer

	International Publishing, 2020. 3. *Classical Mechanics*, TL Chow, CRC press, 2024.

Continuum Mechanics - MTH10428

Module designation	Continuum mechanics
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Bui Xuan Thang
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours. Lectures: 60 hours (in class). Private study: 120 hours (self-study).
Credit points	4 Credits (6.0 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 2A - MTH00013 • Linear Algebra - MTH00030
Module objectives/intended learning outcomes	General Objective: To provide students with fundamental knowledge of continuums in mechanics and mathematical modeling for continuums. Introduce students to in-depth research directions on continuous environmental objects: Deformed solids, Fluids. Specific objectives/course output standards: Knowledge: Understand the concept of continuum, the laws of physics. Apply concepts and laws to mathematically model continuum. Skills: Apply concepts and laws to mathematically model continuums such as elastic solid bodies, fluids.

	Attitude, diligence: After the course, students will understand and be interested in the direction of mathematical research in continuum mechanics.
Content	This course introduces and provides the foundation knowledge for the modules of Deformed Solid Mechanics, Fluid Mechanics, Fracture Mechanics, Composite Materials, and Mechanics Seminar. Analysis of stress Deformation and strain Motion and flow Fundamental laws of continuum mechanics Linear elasticity Fluids
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Tests – process score: 35%. Final exam: 65%. Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Course in Continuum Mechanics* (in Vietnamese), Bui Xuan Thang, lecture note, 2023. 2. *Introductory Continuum Mechanics with Applications to Elasticity*, Tariq Khraishi, Cognella Academic Publishing, 2012. 3. *Continuum Mechanics Modeling of Material Behavior*, Martin H. Sadd, Academic Press, 2019.

Finite Element Method - MTH10429

Module designation	Finite Element Method
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Vu Do Huy Cuong

Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class). Practice: 30 hours (in class). Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: Computational Software Laboratory (MTH00087).
Module objectives/intended learning outcomes	General Objective: To provide basic knowledge of the finite element method. Help students know how to apply the finite element method to solve boundary problems, boundary-value problems appearing in mechanics and physics. Specific objectives/course learning outcomes: Knowledge: Master the concepts and procedures in the finite element method Skills: Know how to apply finite element method to numerically solve boundary problems, boundary-first value problems (from finite element discretization to Matlab programming) Attitude, diligence: Positive and dynamic.
Content	This module includes the following topics: Introduction to numerical methods and finite element methods. Application of the finite element method to scientific and engineering problems. Fundamental theory of the finite element method. Finite element for partial differential equations.

	Finite element for elasticity theory.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Minimum attendance at lectures is 80% (Absences must not exceed 3 times for the entire duration of the lectures). Final score is greater or equal to 5.0/10.0.
Reading list	Main textbook: 1. *Finite Element Method Course Notes* (in Vietnamese), Vu Do Huy Cuong, lecture note, 2023. References: 1. *Finite Element Method Using MATLAB* (in Vietnamese), Nguyen Thoi Trung, Nguyen Xuan Hung, Construction Publishing House, 2015. 2. *MATLAB Codes for Finite Element Analysis*, Antonio J. M. Ferreira, Nicholas Fantuzzi, Amsterdam: Springer Netherlands, 2021. 3. *The Finite Element Method in Engineering*, Rao S.S., Elsevier Inc, 1989.

Complex variable functions - MTH10412

Module designation	Complex Variable Functions
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Ly Kim Ha
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mechanics, AM programme.
Teaching methods	Lectures
Workload (incl. contact hours, self-study hours)	Total: 150 hours Contact hours: lectures 60 hours (in-class).

	Private study: 90 hours (self-study).
Credit points	4 Credits (6.0 ETCS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Analysis 1A (MTH00010) • Analysis 2A (MTH00012) • Analysis 3A (MTH00014)
Module objectives/intended learning outcomes	Equip with basic knowledge of complex numbers, elementary complex functions, and necessary complex analysis knowledge to be able to apply in the specializations of numerical analysis, partial differential equations, digital signal processing, statistics, and probability.
Content	Basis properties of complex numbers and complex functions. Analytic functions, the power expansion, line integral and the theory of residues.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Midterm Exam: 30%, Final Exam: 70%.
Reading list	1. *Complex analysis*, Theodore Gamelin, 2001. 2. *Complex Variables With Applications*, David Wunsch, 2005. 3. *Complex Analysis*, V. Ahlfors, 1966. 4. *Théorie élémentaire des fonctions analytiques d'une ou plusieurs variables complexes*, Henri Cartan, 1961.

Solids Mechanics - MTH10434

Module designation	Solid Mechanics
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Bui Xuan Thang
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mechanics, AM programme

Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 hours. Contact hours: Lecture: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 credits (6 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: • Continuum Mechanics (MTH10428)
Module objectives/intended learning outcomes	General Objective: To provide students with basic concepts of the laws of behavior of solid bodies. Present some basic laws and theorems related to many practical problems of Deformed Solid Mechanics. Provide students with methods of modelling, solving problems and numerical simulation by computers. Knowledge: Students understand the basic concepts of the laws of behavior of solid bodies. Students develop methods of modeling, solving problems and numerical simulation by computers.
Content	The subject plays an important role in providing basic knowledge and opening in-depth research directions in the Deformed Solid Mechanics. The basic knowledge that can be mentioned is the stress-strain relationship in elastic, viscoelastic and elastoplastic theories... In-depth research directions that can be mentioned are the analysis of the behavior of the plate, shell, composite materials, fracture mechanics... This subject inherits a lot of knowledge from the subject of Continuum Mechanics and is also a prerequisite for Finite Element Method, Fracture Mechanics, and Stability and Vibration: Linear elastic solids Formulation and solution strategies Strain energy and related principles Two-dimensional formulation and problem solution

	Anisotropic elasticity Thermoelectricity Nonhomogeneous elasticity Numerical finite element method An introduction of plasticity.
Examination forms	Report, Essay exam
Study and examination requirements	Tests - process score: 35%. Final exam: 65%.
Reading list	1. *Theory of Elasticity* (in Vietnamese), Đao Xuan Bich, VNU Press HaNoi. 2000. 2. *Elasticity: Theory, Applications, and Numerics*, M. H. Sadd, 2020. 3. *Continuum Mechanics Modeling of Material Behavior*, M. H. Sadd, Academic Press, 2019. 4. *Introduction to Nonlinear Finite Element Analysis*, Nam-Ho Kim, Springer Science & Business Media, 2014.

Fluid Mechanics - MTH10435

Module designation	Fluid mechanics
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Nguyen Thi Yen Ngoc
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Mechanics, AM programme
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	Total: 150 Hours Contact hours: Lecture: 60 hours (in class) Private study: 90 hours (self-study)

Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: • Continuum Mechanics (MTH10428)
Module objectives/intended learning outcomes	This course provides fundamental knowledge on fluid mechanics. Knowledge: understanding the fundamental concepts and basic laws of fluid mechanics. Skills & competences: cognitive and practical abilities to use this knowledge in analyzing the physical and mathematical meaning of fluid mechanics problems and solving some basic problems in certain cases.
Content	This module includes the following topics: • Introduction • Fluid statics • Conservation of mass • Inviscid flow • Conservation of momentum • Viscous flow • Conservation of energy
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nguyen Thi Yen Ngoc, *Course in Fluid Mechanics* (in Vietnamese), lecture note, 2022. 2. Munson, Young, Okiishi, and Huebsch, *Fundamentals of Fluid Mechanics*, Wiley, 2009. 3. Fay J.A., *Introduction to Fluid Mechanics*, MIT Press, 1998. 4. Joseph M. Powers, *Mechanics of Fluids*, Cambridge University Press, 2023. 5. Yunus A. Cengel and John M. Cimbala, *Fluid Mechanics: Fundamentals and Applications*, McGraw Hill, 2024.

2.2. ELECTIVE COURSES

Symbolic Programming for Applied Problems - MTH10611

Module designation	Symbolic Programming for Applied Problems
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Vu Do Huy Cuong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study).
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 2A (MTH00013) • Linear Algebra (MTH00030) Recommended prerequisites: • Computational Software Laboratory (MTH00087)
Module objectives/intended learning outcomes	General Objective: To guide learners in using computers to assist in simulating and solving mechanical and real-world problems. Specific objectives/course output standards: Knowledge: After completing the course, students will understand the principles behind certain physical and man-made phenomena in daily life. They will be able to solve (exact solutions) and simulate (using computer software) these phenomena.

	Skills: Computational programming, Apply concepts and laws to mathematically model to real problems. Attitude and diligence: After the course, students will understand and be interested in the computing and simulation of mathematical mechanics for real problems.
Content	This course introduces and provides the foundation knowledge for the modules of Symbolic programming. Symbolic programming with Matlab Kinematics problem Optimization problem Heat transfer problem, wave propagation problem
Examination forms	written exams, lab
Study and examination requirements	Tests – process score: 35% Final exam: 65 %
Reading list	1. *Symbolic Programming with MATLAB for Applied Problems* (in Vietnamese), Vu Đo Huy Cuong, VNU-HCM Press, 2016. 2. *Learn to Code by Solving Problems: A Python Programming Primer*, Daniel Zingaro, No Starch Press, 2021 3. *Physics Problem-Solving Techniques for Understanding and Success in First Year Mechanics*, Christine Berven, Cognella Academic Publishing, 2022.

Mechanics Seminar - MTH10520

Module designation	Mechanics Seminar
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Vu Do Huy Cuong Dr. Bui Xuan Thang

	Dr. Nguyen Thi Yen Ngoc MSc. Le Van Chanh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Project, seminar
Workload (incl. contact hours, self-study hours)	Total workload: 180 hours. Contact hours: Lectures 60 hours. Private study including examination preparation, specified in hours: 120 hours.
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	Successfully completed 124/134 credits.
Module objectives/intended learning outcomes	After completing the course, students will be able to: Apply mathematical principles to solve real-world engineering problems. This includes using calculus, linear algebra, differential equations, and other mathematical tools to analyze forces, stresses, strains, heat transfer, fluid flow, and other physical phenomena. Develop and use computational models to simulate mechanical systems. This involves using software to create virtual models of machines, structures, and other systems, and then running simulations to predict their behavior under different conditions. Design and analyze mechanical systems. This involves applying the principles of mechanics, materials science, and thermodynamics to create new machines, devices, and structures. Present and defend the results of the project at the Students' Seminar Conference.
Content	Various topics
Examination forms	Essay, oral presentation

Study and examination requirements	Minimum attendance at weekly meetings is 80%. Final defense is required. Final score is greater or equal to 5.0/10.0
Reading list	Diverse documents depend on the instructor.

3. SPECIALIZATION IN DIDACTICS AND METHODOLOGY OF TEACHING MATHEMATICS

3.1. COMPULSORY COURSES

Pedagogical Psychology - MTH10101

Module designation	Pedagogical Psychology
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Tran Huong Thao
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	270 Hours Contact hours: Lectures: 30 hours (in class); 60 practical hours. Private study: 180 hours (self-study)
Credit points	4 Credits (7 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course equips students with basic psychological knowledge about teachers and learners, problem-solving skills, teamwork skills and applies them to teaching. Knowledge: Research the psychological problems of managing the teaching-learning process. Determine the conditions to ensure intellectual development in the teaching-learning process. Describe the formation and cultivation of the teacher's personality as well as the psychological development of the learner. Skills & competences: Developing teamwork skills, presentation skills, report writing, and pedagogical

	communication skills through various approaches tailored to different age groups of the target audience being taught.
Content	This module includes the following topics: General overview. Fundamental issues in educational psychology Personality psychology of teachers. Pedagogical activities according to developmental psychology.
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Le Van Hong, Le Ngoc Lan and Nguyen Van Thang, *Age Psychology and Pedagogical Psychology*, Hanoi, 1995. 2. Hoang Anh & Vu Kim Thanh, *Pedagogical Communication*, Hanoi, 1995. 3. Tran Thi Huong, Nguyen Thi Bich Hanh, Ho Van Lien & Ngo Dinh Qua, *General Pedagogy*, Ho Chi Minh City University of Education, 2009. 4. Tran Thi Huong, Vo Thi Bich Hanh, Ho Van Lien, Vu Thi Sai, Vo Thi Hong Truoc, *General Education*, Ho Chi Minh City University of Education, 2009. 5. Nguyen Xuan Thuc (editor-in-chief), *General Psychology*, Ho Chi Minh City University of Education, 1995. References: 1. Peter Filene, The Joy of Teaching: *A Practical Guide for New College Instructors*, the University of North Carolina Press, 2005. 2. Ken Bain, *What the Best College Teachers Do*, Harvard University Press, 2004. 3. Rainer Zwisler, Einführung in die Pädagogische Psychologie, 4. Auflage, Weinheim und München, *Psychologische Verlags Union Beltz*, 1994.

	4. J. Zumbach und H. Mandl (Hrsg.), *Paedagogische Psychologie in Theorie und Praxis*, Verlag Hogrefe, 2008. 5. Klafi, W., *Studien zur Bildungstheorie und Didaktik*, Weinheim: Beltz, 1975. 6. Rainer Zwisler, Einführung in die Pädagogische Psychologie, Auflage, Weinheim und München, *Psychologische Verlags*, Union Beltz, 1994. 7. Watzlawik, P., Die. *5 Kommunikationsaxiome. In: Kommunikation und Selbstsicherheit, Interaktionsspiel für Schule*, Jugendarbeit und Erwachsenenbildung, Muelheim, 1995.

Optimal Teaching and Learning Methods - MTH10102

Module designation	Optimal Teaching and Learning Methods
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Phan Nguyen Ai Nhi MSc. Chau Thi Hieu
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme. Elective for others programme.
Teaching methods	Lecture, project-based learning
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 30 hours (in class); 30 exercise hours Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	This course plays a crucial role in nurturing foundational knowledge of the fundamental characteristics of contemporary education. It also introduces positive teaching methods (PPGDTC) from the concept, organization, and advantages and disadvantages of each teaching method. From here, learners can approach, select, and flexibly apply these teaching methods in practical classroom settings, teaching in different environments through project activities, in other courses, during teaching internships, and in subsequent teaching activities. Knowledge: Understanding various teaching methods and current teaching issues. Overview of planning and time management. Skills & competences: Evaluating the application of methods in real teaching situations. Developing skills in teamwork, presentation, report writing, planning, and time management.
Content	This module includes the following topics: Project-based learning Community Service Learning Positive Teaching Methods
Examination forms	Midterm and final exam: project and written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Robert J. Marzano, Debra J. Pickering, Jane E. Pollock. *Effective Teaching Strategies*. (Vietnamese version) Vietnam Education Publishing House, 2013. 2. *Northern Ireland Curriculum Active Teaching and Learning Methods for Key Stages 1, 2 &3*, APMB Publication, 2007. 3. Meg. Stephenson, *Service Learning in the Curriculum*, University of Utah, USA, 2003. 4. G. Stewart, *Promoting & Managing Effective Collaborative Group Work*, Belfast Education and Library Board,2014.

	References: 1. Various authors. *Current Issues in Education – Perspectives and Solutions*. Tri Thuc Publishing House, 2008. 2. Dang Thanh Hung. *Teacher–Student Interaction in Classroom Activities* (in Vietnamese). Vietnam Education Publishing House, 2007. 3. Nguyen Hien Le. *Self-Study – A Necessity of the Era* (in Vietnamese). Culture and Information Publishing House, 2007. 4. Barbara Gross David, *Tool of Teaching*, Jossey-Bass, 2009. 5. James M. Banner. Jr. & Harold C. Cannon, *The Elements of Teaching*, Yale University Press 1997. 6. Jeffrey S. Lantis, Lynn M. Kuzma, John Boehrer, *The New International Studies Classroom - Active Teaching*, Active Learning, Lynne Rienner Publishers, United State, 2000. 7. Robert S. Feldman, *Power Learning*, McGraw Hill, USA 2003. 8. Debra H. Hydorn, *Community Service Learning in Mathematics*, 2011.

General Education - MTH10104

Module designation	General Education
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc. Phan Nguyen Ai Nhi MSc. Chau Thi Hieu
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme. Elective for others programme.
Teaching methods	Lecture, project-based learning

Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 30 hours (in class) + 30 exercise hours Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course covers issues in general education, learning outcomes in knowledge, educational objectives regarding attitudes, external orientation attitude groups, internal orientation attitude groups, educational theory, and an overview of educational pathways. Knowledge: Comparing the educational model in Vietnam with models in other parts of the world. Analyzing the relationship between educational theory and practical issues. Skills & competences: Selecting teaching methods, assessment methods to design and deliver detailed lesson plans.
Content	This module includes the following topics: Education and the Nature of Education Understanding Human Development Designing Educational Programs Learning Outcomes in Knowledge Educational Objectives in Attitudes Sharing Adaptation Defense Effort Stability Intelligence

	Common Issues in Educational Theory Content and Behavior Rules Evaluation Lesson Design (Basic).
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Nguyen Thi Bich Hong & Vo Van Nam. *General Pedagogy. Textbook*, Ho Chi Minh City University of Education, 2004. References: 2. Nguyen An. *General Pedagogy. Textbook*, Ho Chi Minh City University of Education, 1997. 3. Ha The Ngu & Dang Vu Hoat. *Pedagogy,* Vol. 1. Vietnam Education Publishing House, 1988. 4. Dang Vu Hoat (chief editor). *Pedagogy*, Vol. 2. Vietnam Education Publishing House, 1995. 5. Pham Viet Vuong. *Pedagogy*. Vietnam National University Press Hanoi, 2000. 6. Nguyen Thi Bich Hanh & Tran Thi Huong. *Theory of Instruction (Didactics) Textbook*, Ho Chi Minh City University of Education, 2004. 7. Peter Filene, *The Joy of Teaching*, University of North Carolina Press, 2005. 8. Ken Bain, *What the best college teachers do*, Harvard University Press, 2004. 9. E. F. Crawley, J. Malmqvist, S. Oslund, D. R. Brodeur, *Rethinking Engineering Education: The CDIO Approach*. Springer Science+Business Media, 2007.

Didactics of Teaching - MTH10105

Module designation	Didactics of Teaching
Semester(s) in which the module is taught	6th semester

Person responsible for the module	MSc. Be Phuong Thao
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme.
Teaching methods	Lecture, project-based learning
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 30 hours (in class) Practice: 30 (in class) Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The theory of teaching presents fundamental concepts in the process of imparting and acquiring knowledge in general, particularly within the school context. Simultaneously, it describes and explains phenomena related to the relationship between teaching and students' learning experiences. Knowledge: teaching principles, principles of knowledge transmission. Skills & competences: Handling various teaching situations. Designing teaching scenarios, being proactive, having problem-solving skills, adaptability, and self-regulation.
Content	This module includes the following topics: Fundamental Concepts System of Teaching Principles Principles of Knowledge Transmission
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.

Reading list	Main textbook: 1. Dang Duc Trong, Do Thi Bich Tram, *Didactics*, Vietnam National University – Ho Chi Minh City Publishing House, 2014. 2. Duong Thi Kim Oanh, *Didactics and Teaching Methods*, Vietnam National University – Ho Chi Minh City Publishing House, 2022. References: 1. Bessot, A.; Comiti, C.; Chau, L. T. H.; Tien, L. V. *Basic Elements of Mathematics Didactics*. Vietnam National University – Ho Chi Minh City Publishing House, 2007. 2. Marzano, R. J.; Marzano, J. S.; Pickering, D. J. *Effective Classroom Management*. Vietnam Education Publishing House, 2011 (Vietnamese version). 3. Thai Duy Tuyen, Teaching Methods: *Tradition and Innovation*, Vietnam Education Publishing House, 2010. 4. Nguyen Huu Chau, *Fundamental Issues in Curriculum and the Teaching Process*, Vietnam Education Publishing House, 2005.

Elementary Number Theory and Mathematical Logic - MTH10106

Module designation	Elementary Number Theory and Mathematical Logic
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Tran Nam Dung
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours:

	Lectures: 30 hours (in class) Practice: 30 hours (in class) Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course provides students with fundamental knowledge in mathematics and mathematical logic, which helps them to understand and solve mathematical problems in the general education curriculum. It also cultivates their skills in rigorous reasoning for mathematical proofs and enhances their ability to articulate issues succinctly, coherently, and with adherence to standards. Knowledge: get familiar with basic concepts elementary number theory and mathematical logic. Skills & competences: cognitive and practical abilities to use these knowledges in solving mathematics problems and teaching mathematics.
Content	This module includes the following topics: 1. Fundamental Concepts 2. Congruence and Congruence Equations 3. Mathematical Functions 4. Diophantine Equations 5. Propositions and Predicates 6. Mathematical Induction, Contradiction Method
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Kenneth H. Rosen. *Discrete Mathematics and Its Applications*. 8th ed., McGraw-Hill Education, 2019. 2. Joseph H. Silverman. *A Friendly Introduction to Number Theory*. 4th ed., Pearson, 2023.

	References: 1. G. Polya, *Mathematics and Plausible Reasoning*, Vietnam Education Publishing House, 1999. 2. Hoang Chung, *Number Theory - The Queen of Mathematics*, Vietnam Education Publishing House, 1997.

Methods of Teaching Mathematics 1- MTH10110

Module designation	Methods of Teaching Mathematics 1
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Pham Hoang Hai MSc. Nguyen Do Truong An
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 30 hours (in class) Practice: 30 hours (in class) Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The course addresses issues related to propositions, sets, functions, equations, algebraic inequalities, inequalities, limits, continuous functions, and the derivatives of functions.

	- Knowledge: Overview of fundamental algebraic mathematics knowledge. Recognizing current issues in teaching. Describing methods of teaching mathematics. - Skills & competences: cognitive and practical abilities to use this knowledge in solving mathematics problems. Selecting appropriate tools for teaching mathematics. Formulating ideas for organizing new teaching approaches suitable for each student group
Content	This module includes the following topics: 1. Propositions-set-function, 2. Equations, Inequalities and systems of equations, Equations, 3. Inequalities involving absolute value, Equations, 4. Inequalities involving square roots, Inequalities, 5. Combinatorics 6. Probability 7. Sequences 8. Arithmetic progressions, 9. Limits of functions, continuous functions, 10. Derivatives of functions
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Tran Nam Dung (General Editor), Tran Duc Huyen (Chief Editor) & colleagues. *Mathematics 10, Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2022. 2. Tran Nam Dung (General Editor), Tran Duc Huyen (Chief Editor) & colleagues. *Mathematics 11*, *Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2023. 3. Tran Nam Dung (General Editor), Tran Duc Huyen (Chief Editor) & colleagues. *Mathematics 12, Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2024. References:

	1. Hoang Thi Thanh Liem, Nguyen Thi Ninh, Nguyen Van Quyet, Vu Thu. *Elementary Problem-Solving Practice in Mathematics, Vol. 1*. Vietnam Education Publishing House, 1986.

Methods of Teaching Mathematics 2- MTH10111

Module designation	Methods of Teaching Mathematics 2
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Pham Hoang Hai MSc. Nguyen Do Truong An
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 30 hours (in class); 30 practical hours Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course equips students with the ability to solve, classify exercises, present textbooks, and to be able to teach high-school Mathematics. Knowledge: Provide an overview of fundamental geometric mathematics knowledge.

	Skills & competences: Apply teaching and learning methods and educational psychology to the organization and management of classroom processes
Content	This module includes the following topics: 2. Methods of preparation of lesson plan and exercises 3. The scalar product and applications 4. Coordination method in plane 5. Transformation and similarity in planes 6. Parallel in space 7. Vector in space, perpendicular 8. Polyhedron and their volumes 9. Spheres, Cylinder, Cones
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Tran Nam Dung (General Editor), Tran Duc Huyen (Chief Editor) & colleagues. *Mathematics 10, Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2022. 2. Tran Nam Dung (General Editor), Tran Duc Huyen (Chief Editor) & colleagues. *Mathematics 11, Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2023. 3. Tran Nam Dung (General Editor), Tran Duc Huyen (Chief Editor) & colleagues. *Mathematics 12, Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2024. 4. Tran Duc Huyen (Chief Editor) & colleagues. *Workbook - Mathematics 10, Volumes 1 - 2*. Creative Horizons Series. Vietnam Education Publishing House, 2022. References: 1. Van Nhu Cuong, Pham Khac Ban, Ta Man. *Geometry Workbook, Grade 11*. Vietnam Education Publishing House, 2007.

	2. Van Nhu Cuong, Pham Khac Ban, Le Huy Hung, Ta Man. *Geometry Workbook, Grade 12*. Vietnam Education Publishing House, 2008. 3. *Teacher-Training Manual for Implementing the Grade 10 Upper-Secondary Curriculum and Textbook – Mathematics*. Vietnam Education Publishing House, 2006. 4. *Teacher-Training Manual for Implementing the Grade 11 Upper-Secondary Curriculum and Textbook – Mathematics*. Vietnam Education Publishing House, 2006. 5. *Guidelines for Implementing the Grade 12 Curriculum and Textbook – Mathematics*. Vietnam Education Publishing House, 2008.

Pedagogical Practice - MTH10112

Module designation	Pedagogical Practice
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Dinh Thi Kim Lien
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Didactics and Methodology of Mathematics Teaching, AM programme. Elective for others programme.
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	360 Hours Contact hours: Lectures: 120 practical hours (in class) Private study: 240 hours (self-study)
Credit points	4 Credits (8 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Pedagogical Psychology (MTH10101) • General Education (MTH10104)

	• Didactic of Teaching (MTH10105) • Methods of Teaching Mathematics 1 (MTH10110) • Methods of Teaching Mathematics 2 (MTH10111)
Module objectives/intended learning outcomes	The course covers the internship process in pedagogy, specifically focusing on the responsibilities of homeroom activities and teaching at secondary schools. It involves organizing teaching and classroom management, activities related to lesson planning, and writing internship reports to create conditions for students to acquire extensive knowledge, skills, and practical experience during the internship. Knowledge: Organizing teaching and classroom management, activities related to lesson planning, and writing internship reports, practical experience. Skills & competences: Developing and enhancing teaching skill.
Content	This module includes the following topics: General overview of the pedagogical practice process Homeroom teacher role practice Teaching practice
Examination forms	Midterm and final exam: report final
Study and examination requirements	Overall grade: minimum 5.0/10.0.
Reading list	1. Ho Chi Minh City University of Education. *Regulations on Professional Teaching Practice in Credit-Based Teacher Education*. 2019. 2. Ministry of Education and Training. *Regulations on Teaching Internship*. 2019. 3. Ministry of Education and Training. *Circular No. 12/2021/TT-BGDĐT: Program and Implementation of Pedagogical Training for Prospective Secondary and High-School Teachers*. 2021.

3.2. ELECTIVE COURSES

Quality and Quality Management - MTH10103

Module designation	Quality and Quality Management

Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Dinh Thi Kim Lien
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 30 hours (in class), 30 practical hours Private study: 120 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	The course addresses issues related to quality, quality management, total quality management, continuous improvement through Kaizen, applied to enhancing teaching through lesson research Knowledge: Identifying practical education issues and methods to enhance teaching quality. Skills & competences: Developing and enhancing presentation skills, scientific report writing and analyzing data.
Content	This module includes the following topics: 1. Quality and Customers. 2. Total Quality Management (TQM). 3. Culture and cultural scripts. 4. Continuous improvement through Kaizen. 5. Continuous improvement in choices.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.

Reading list	Main textbook: 1. Japan Human Relations Association, *Guiding continuous improvement through employee suggestions*, 1997 2. NCafÉ. *A Roadmap for Quality Transformation in Education: A Guide for Local Education Reform Leaders*. NC Awards for Excellence Press, 2020 3. Jeronen, E. (Ed.). (2021). *Transitioning to Quality Education: A Guide for Sustainable Education and SDG-Driven Quality*. MDPI. References: 1. Business Edge. *Understanding Quality*. 2003. 2. Business Edge. *Achieving Quality*. 2003. 3. Business Edge. *Assessing Quality*. 2003. 4. E. F. Crawley, J. Malmqvist, S. Oslund, D. R. Brodeur, *Rethinking Engineering Education: The CDIO Approach*, 2007.

Teaching Mathematics in English 1 - MTH10119

Module designation	Teaching Mathematics in English 1
Semester(s) in which the module is taught	Both 6th and 7th semester
Person responsible for the module	Dr. Ta Thi Nguyet Nga
Language	English
Relation to curriculum	Elective
Teaching methods	Lecturing, discussion, think-pair-share, Project-based learning,
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 credits (3 ECTS)

Required and recommended prerequisites for joining the module	Proficiency in both mathematics and English. Familiarity with teaching strategies and classroom management.
Module objectives/intended learning outcomes	This course is designed to equip students with the skills and strategies needed to effectively teach mathematics using English as the medium of instruction. Participants will explore techniques for creating a language-rich math classroom, fostering mathematical communication, and supporting English language learners in understanding and solving mathematical problems. The course aims to empower students to facilitate a seamless integration of mathematics and language learning. By the end of this course, students should be able to: Understand the importance of the structure and concepts in teaching mathematics in English. Develop strategies for explaining mathematical concepts clearly in English. Promote mathematical communication and language development. Adapt instructional materials to support English language learners. Create a language-rich math classroom environment. Evaluate the effectiveness of English-medium math instruction.
Content	Operation Review Algebra Basic Geometry Trigonometry Probability and Statistics Calculus
Examination forms	Class Participation and Engagement: 10%

	Assignments: 20% Midterm Exam: 30% Final project: 40%
Study and examination requirements	None
Reading list	1. Robert F. Blitzer, *Algebra and Trigonometry*, Pearson 2022. 2. Karen Morrison, Nick Hamshaw, and Dea Susanto. *Cambridge IGCSE® Mathematics Core and Extended Coursebook*. 3rd Edition, Cambridge University Press, 2023. 3. Sullivan, Michael & Kathleen Miranda. *Calculus for the AP® Course*. 4th ed., W. H. Freeman, 2023.

Teaching Mathematics in English 2 - MTH10120

Module designation	Teaching Mathematics in English 2
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Dr. Ta Thi Nguyet Nga
Language	English
Relation to curriculum	Elective
Teaching methods	Lecturing, discussion, think-pair-share, Project-based learning,
Workload (incl. contact hours, self-study hours)	90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ECTS)
Required and recommended prerequisites for joining the module	Proficiency in both mathematics and English. Familiarity with teaching strategies and classroom management.

Module objectives/intended learning outcomes	This course is designed to equip students with the skills and strategies needed to teach mathematics effectively using English as the medium of instruction. Learners will explore methods for creating a language-rich learning environment, fostering mathematical communication, and supporting students in developing mathematical thinking through English. The course focuses on the seamless integration of mathematical content and language skills, enabling educators to deliver clear, comprehensible, and effective instruction. By the end of this course, students should be able to: Understand and analyze the linguistic and cognitive aspects of teaching mathematics in English, as well as the challenges students may face. Develop effective teaching strategies for explaining mathematical concepts in English with clarity, accuracy, and appropriateness for different student groups. Encourage mathematical communication by using precise language to explain, discuss, and present mathematical ideas Select and adapt instructional materials to accommodate students with varying levels of English proficiency while maintaining mathematical rigor and depth Create a language-rich classroom environment that fosters mathematical thinking through discussions, collaboration, and presentations in English. Assess and refine the effectiveness of English-medium math instruction based on student feedback and appropriate pedagogical criteria.
Content	General of Teaching math in English Teaching Algebra Teaching Geometry Teaching Probability and Statistics Teaching Calculus

Examination forms	Class Participation and Engagement: 10% Assignments: 20% Midterm Exam: 30% Final project: 40%
Study and examination requirements	None
Reading list	1. Stewart, James; Daniel K. Clegg; Saleem Watson. *Calculus: Early Transcendentals*. 9th ed. Brooks/Cole, Cengage Learning, 2021. 2. Sullivan, Michael & Kathleen Miranda. *Calculus for the AP® Course*. 4th ed., W. H. Freeman, 2023. 3. Goldie, Sophie. *Cambridge International AS & A Level Mathematics: Pure Mathematics 1*. 2nd ed., Hodder Education, 2022 (digital edition).

Elementary Algebra – MTH10121

Module designation	Elementary Algebra
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Ta Thi Nguyet Nga
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	None

Module objectives/intended learning outcomes	This course addresses issues related to methods and strategies for solving problems, equations and systems of equations, inequalities and extremum conditions, combinatorial algebra, and methods for solving problems related to sequences and series. Knowledge: Overview of foundational knowledge in algebra. Skills & competences: Selecting appropriate theoretical foundations to solve corresponding exercises. Proactively updating and improving new teaching methods.
Content	This module includes the following topics: 1. Methods and strategies for problem solving. 2. Equations and systems of equations 3. Inequalities and extremum 4. Combinatorial algebra 5. Methods for solving problems of sequences and series.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Doan Quynh, *Higher-Level Algebra 10*, Vietnam Education Publishing House, 2006 2. Doan Quynh, *Higher-Level Algebra and Analysis 11*, Vietnam Education Publishing House, 2007 3. Doan Quynh, *Higher-Level Analysis 12*, Vietnam Education Publishing House, 2008 References: 1. Doan Quynh, *Mathematics for Gifted Students – Algebra 10*, Vietnam Education Publishing House, 2009 2. Doan Quynh, *Mathematics for Gifted Students – Algebra and Analysis 11*, Vietnam Education Publishing House, 20100. 3. Doan Quynh, *Higher-Level Analysis 12*, Vietnam Education Publishing House, 2011

Elementary Geometry - MTH10122

Module designation	Elementary Geometry
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Tran Nam Dung
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Project, seminar
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course equips students with skills in analysis, geometric problem-solving, methods, and problem-solving strategies Knowledge: important knowledge to solve geometry problems. Skills & competences: cognitive and practical abilities to use this knowledge in solving geometry problems.
Content	This module includes the following topics: 1. Methods and Strategies for Problem Solving 2. Solving Problems using Vector Methods 3. Solving Problems using Transformations 4. Coordinate Methods in Plan 5. Coordinate Methods in Space 6. Methods for solving three-dimensional geometry problems

Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Higher-Level Geometry 11, *Ministry of Education and Training*, Vietnam Education Publishing House, 2020. 2. Nguyen Van Hoa, *Systematic Knowledge of Geometry 10 - 11 - 12*, Ho Chi Minh City University of Education Publishing House, 2021. References: 1. Doan Quynh (Chief Editor), *Mathematics for Gifted Students - Geometry 10*, Vietnam Education Publishing House, 2009. 2. Doan Quynh (Chief Editor), *Mathematics for Gifted Students - Geometry 11*, Vietnam Education Publishing House, 2010. 3. Doan Quynh (Chief Editor), *Mathematics for Gifted Students - Geometry 12*, Vietnam Education Publishing House, 2011.

Application of Advanced Mathematical Methods in Solving Complex High School's Mathematics Problem - MTH10123

Module designation	Application of Advanced Mathematical Methods in Solving Complex High School's Mathematics Problem
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Ta Thi Nguyet Nga
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours:

	Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Equip students with advanced mathematical knowledge through various applications to solve problems in the high school curriculum. This helps students reinforce and enhance their advanced mathematical knowledge, while also teaching them how to solve everyday problems using modern and elegant tools. Knowledge: advanced mathematical knowledge in solving problems Skills & competences: solving everyday problems using modern and elegant tools.
Content	This module includes some of the following topics: (change over year) 1. Applications of Mathematical Analysis 2. Applications of Advanced Geometry 3. Applications of Graph Theory 4. Applications of Group Theory 5. Applications of Symmetric Polynomials
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Stephen Siklos. *Advanced Problems in Mathematics: Preparing for University*. Expanded ed., Open Book Publishers, 2020. 2. Arthur Engel. *Problem-Solving Strategies* (Problem Books in Mathematics). Corrected ed., Springer, 1997 - reissued 2021. References:

	1. Dang Hung Thang, *Some Fundamental Concepts of Finite Graphs*, Education Publishing House, 2004 2. Ha Huy Khoai, *Enrichment Topics in High-School Number Theory*, Education Publishing House, 2006 3. J. Rotman, *An introduction to the theory of group*, Springer - Verlag, 1999. 4. K.H. Rosen, *Discrete Mathematics and Its Aplications*, Addison - Wesley Publishing Company, 1993.

Classroom Management and Organization - MTH10124

Module designation	Classroom Management and Organization
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Phan Nguyen Ai Nhi MSc. Chau Thi Hieu
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lab work, project
Workload (incl. contact hours, self-study hours)	270 Hours Contact hours: Lectures: 30 hours (in class) Practice: 60 hours (in class) Private study: 180 hours (self-study)
Credit points	4 Credits (7 ECTS)
Required and recommended prerequisites for joining the module	Pedagogical Psychology (MTH10101)
Module objectives/intended learning outcomes	The course provides an overview of issues related to classifying learners, the adaptability between teaching methods and assessment (TM&A) for each learner category. It particularly emphasizes the effective

	organization and management of the classroom, preparing future teachers with the necessary skills for learners. Knowledge: Overview of the foundational knowledge of the process of classroom organization and management. Presents the psychophysiological characteristics of each age group of students. Classifies and interacts effectively with each group of students. Skills & competences: Analyze the effectiveness of teaching methods. Select appropriate assessment methods for each specific content and audience. Cultivate the ability to work independently, study, and conduct comprehensive research
Content	This module includes the following topics: Classify learner groups, Adaptability between teaching methods and assessment (TM&A) for each learner group, Organize and manage the classroom
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Pham Thi Thanh Bình. *Classroom Organization and Management - Theory and Practice*, Vietnam Education Publishing House, 2021. 2. Phan Nguyen Ai Nhi. *Classroom Organization and Management*, 2014. 3. Dang Thanh Hung, *Teacher - Student Interaction in Classroom Activities*, Vietnam National University Press, Ha Noi, 2007. References: 1. Jeffrey S. Lantis, Lynn M. Kuzma, John Boehrer. *The New International Studies Classroom - Active Teaching*, Active Learning, Lynne Rienner Publishers, 2000

	2. Klafi, W., *Studien zur Bildungstheorie und Didaktik*, 1975 3. Meyer, H. L., *Trainings programm zur Lernziel analysis*, 1976 4. Research Council, *How People Learn: Brain, Mind, Experience, and School: Expanded Edition National*, 2000 5. Peter Filene, The Joy of Teaching: *A Practical Guide for New College Instructors*, University of North Carolina Press, 2005.

Classroom Assessment Techniques – MTH10125

Module designation	Classroom Assessment Techniques
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Phan Nguyen Ai Nhi MSc. Chau Thi Hieu
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, lab work, project
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 30 hours (in class) Practice: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	3 Credits (5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	This course is oriented towards researching the roles, characteristics, principles, and forms of assessment, as well as the techniques and tools of assessment. It also focuses

	on establishing standardized evaluation systems to support the future teaching process of learners. Knowledge: Recognizing the roles of teaching and assessment. Gaining an overview of assessment techniques and tools. Skills & competences: Constructed appropriate assessment rubrics, compiled exams compatible with course learning outcomes, applied flexible teaching methods to organize classes.
Content	This module includes the following topics: General Overview Assessment Techniques Rubric - an Effective Assessment Tool Organizing Assessment in Teaching.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Thomas A. Angelo & Todd D. Zakrajsek, *Classroom Assessment Techniques: Formative Feedback Tools for College and University Teachers*, Jossey-Bass (Wiley), 3rd ed., June 26, 2024 2. Linda Suskie, *Assessing Student Learning*, Jossey-Bass, 2009 References: 1. Barbara Gross David, *Tools for Teaching*, Jossey-Bass, 2009 2. Mensa, *Keep Your Brain Fit*, Carlton Books Ltd, 2007 3. National Research Council, *How People Learn: Brain, Mind, Experience and School*, National Academies Press, 2000 4. Robert S. Feldman, *Power Learning*, McGraw-Hill, 2003

	5. Steve Frankland, *Enhancing Teaching and Learning through Assessment*, Springer, 2000.

Pedagogy Seminar - MTH10126

Module designation	Pedagogy Seminar
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Tran Nam Dung MSc. Phan Nguyen Ai Nhi MSc. Chau Thi Hieu Dr. Ta Thi Nguyet Nga MSc. Tran Huong Thao
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Seminar, project
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits / 6 ECTS
Required and recommended prerequisites for joining the module	General Education (MTH10104)
Module objectives/intended learning outcomes	After completing this course, students will be able to identify issues related to education and mathematics, classify the learning environment, assess influencing factors, and apply appropriate teaching methods and assessment techniques. Additionally, they will analyze core factors in the new educational direction, develop teaching skills in accordance with the designed curriculum,

	and engage in discussions on specialized topics throughout each week. Knowledge: List of the fundamental factors of the educational curriculum and mathematics. Determine the learning outcomes, teaching methods, and assessment techniques. Classify the learning environment and influence factors. Provide an overview of new educational perspectives and directions. Skills & competences: Foster a sense of self-directed learning. Cultivate community service awareness.
Content	This module includes the following topics: Overview of Building a Positive Learning Environment Three Specialized Topics (Students may propose topics for research and reporting on their own) Teaching: 3 Specialized Topics Teaching according to Gregorc/Anthony perspectives and the approaches of education innovators worldwide. Teaching with the CDIO approach. STEM/STEAM Education. Assessment: 2 Specialized Topics The influence of assessment in the teaching-learning process and proposed recommendations. Building a Rubric system for assessing teaching and learning activities. Mathematics: 3 Specialized Topics How to help students excel in mathematics? Solutions to support students with disabilities in learning mathematics/developing critical thinking. Building a repository of scenarios/games/model systems, etc., to support the teaching and learning of mathematics.
Examination forms	Midterm and final exam: written exams.

Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Barbara Oakley, Terrence Sejnowski & Alistair McConville. *Learn Like a Pro: Science-Based Tools to Master Any Subject*. Vietnamese edition translated by Lien Huong. AlphaBooks & The Gioi Publishing House, 2022. 2. Nguyen Tien Dung. *How to Learn and Teach Mathematics*. Sputnik Books, 2015. 3. John A. Van de Walle, Karen S. Karp & Jennifer M. Bay-Williams. *Elementary and Middle School Mathematics: Teaching Developmentally*. 11th ed., Pearson, 2023 4. National Academies of Sciences, Engineering, and Medicine. *Scaling and Sustaining Pre-K-12 STEM Education Innovations: Systemic Challenges, Systemic Responses*. National Academies Press, 2025. References: 1. Anne Bayetto, *Teaching students with learning difficulties in Mathematic*, SPELD SA, 2015 2. James Bellanca, *200+ Active Learning Strategies and Projects*, Corwin Press, 2009 3. Giselle O., Martin-Kniep, *Become A Better Teacher*, ASCD, 2000 4. Ken Bain, *What the Best College Teachers Do*, Harvard University Press, 2004 5. Robert J. Marzano, *Classroom Management that Works*, ASCD, 2003 6. Peter Filene, *The Joy of Teaching: A Practical Guide for New College Instructors*, The University of North Carolina Press, 2005 7. Steve Frankland, *Enhancing Teaching and Learning through Assessment*, Springer Netherlands, 2010, 8. Linda Suskie, *Assessing Student Learning*, Jossey-Bass, 2009. 9. Thomas A., Angelo, K., Patricia Cross, *Classroom Assessment Techniques*, Jossey-Bass, 1993.

4. SPECIALIZATION IN MATHEMATICAL FINANCE

4.1. COMPULSORY COURSES

Elementary Financial Mathematics - MTH10201

Module designation	Elementary Financial Mathematics
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Phan Thi Phuong
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study).
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 1A (MTH00011) • Calculus 2A (MTH00013)
Module objectives/intended learning outcomes	Equip students with the basic knowledge of finance and financial mathematics for discrete non-random models.

Content	Including the theory of interest rates, money chains, forms of borrowing, appraisal of investment projects, valuation of bonds and stocks.
Examination forms	Midterm and Final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Thanh, D. N., Hong, P. T. T., Trong, D. D. (2011). *Basic Financial Mathematics*. (in Vietnamese). VNUHCM Press. 2. Tapiero Charlas (2004). *Risk and financial management: Mathematical and Computational Methods*. Wiley. 3. Chris Ruckman, Joe Francis (2005). *Financial Mathematics, A Practical Guide for Actuaries and other Business Professionals*, 2nd edition, BPP Professional Education, Inc. 4. Claude Anne (2017). *Mathematiques Financieres*, Ellipses Edition Marketing S.A.

Forecasting - MTH10202

Module designation	Forecasting
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Dr. Nguyen Tien Dat
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lectures, exercises, practice sessions

Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Practice: 30 hours (in lab) Private study: 95 hours (self-study).
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Calculus 1A (MTH00011) • Calculus 2A (MTH00013) • Mathematical Statistics (MTH10404)
Module objectives/intended learning outcomes	Provide knowledge and skills to build basic forecasting models, especially with applications in the financial field. Additionally, using statistical software R to build statistical models for forecasting problems.
Content	Introduction to forecasting. Forecasting with linear regression models: Simple linear regression models Multiple linear regression models Forecasting. Forecasting with time series models including: Introduction to the time series models. Exponential smoothing models ARMA models SARIMA models
Examination forms	Practice exercises (30%). Midterm exam (30%): written exams.

	Final exam (40%): projects.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Davis Richard A., Brockwell Peter J. *Introduction to Time series and forecasting*, Springer, 2016. 2. Shumway Robert H., Stoffer David S. *Time series analysis and its applications with R examples*, Springer, 2017. 3. R. J. Hyndmand, G. Athanasopoulos (2021). *Forecasting: principles and practice*. OTexts.org.

Mathematical Finance Models - MTH10203

Module designation	Mathematical Finance Models
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Nguyen Dang Minh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures 45 hours (in class) Practice: 30 hours (in lab) Private study: 90 hours (self-study).
Credit points	4 credits (6.5 ECTS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Knowledge: Consolidate basic and advanced knowledge of probability theory to apply to solving problems in insurance and finance. Consolidate knowledge about financial markets and financial instruments as the foundation for building financial models to solve practical problems. Basic introduction to decision making theory. Building financial risk management models. Set up several pricing models. Skills: Applying theory to build models suitable to reality, reflecting the nature of relationships in business-finance. Using computer software to support calculations in the subject. Study attitude: attend school fully and do homework, ensure self-study time at home.
Content	This course introduces basic financial models to help students understand and apply financial knowledge to solve fundamental financial problems such as decision making, risk assessment, and valuation. … in a way that has a clear scientific basis.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook:

	1. Oliver Linton, *Financial Econometrics: Models and Methods*, Cambridge University Press, 2019. References: 2. James R. Hitchner, *Financial Valuation: Applications and Models*, + Website, Wiley, 2017. 3. Thomas A Severini, *Introduction to Statistical Methods for Financial Models*, Chapman and Hall/;CRC Press, 2019. 4. John S. Tjia, *Building Financial Models, Third Edition: The Complete Guide to Designing, Building, and Applying Projection Models*, 3rd, McGraw Hill Professional, 2022. 5. Avishek Nag, *Stochastic Finance with Python: Design Financial Models from Probabilistic Perspective*, Xpress, 2024.

Financial and Monetary Theory - MTH10204

Module designation	Financial and Monetary Theory
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Nguyen Huu Toan MSc. Phan Thi Phuong
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class)

	Exercises: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General Objective: This course explores issues related to interest rates, and how financial markets and financial institutions work. This course is divided into 5 parts. Part 1 provides an overview of the financial system, how interest rates are calculated, and how interest rates are structured. Part 2 explores how the debt securities market works. Part 3 learns about the stock market. Part 4 explores the derivatives market. Section 5 explores the activities of commercial banks and non-banks.
Content	Overview of the financial system and financial institutions. Debt stock market. Market share. Derivatives stock market. Commercial banks and non-banking organizations.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Frederic Mishkin, Stanley Eakins. *Financial Markets and Institutions*, Global Edition, 10th, Pearson, 2023. References: 1. Jeff Madura. *Financial Markets and Institutions*, 11th, Cengage Learning, 2014. 2. Stephen A. Ross. *Fundamentals of Corporate Finance*, 13th, McGraw-Hill Education, 2022.

	3. Anthony Saunders, Marcia Millon Cornett. *Financial Markets and Institutions*, 5th, McGraw-Hill/Irwin, 2011.

Advanced Financial Mathematics - MTH10209

Module designation	Advanced Financial Mathematics
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Prof. Dang Duc Trong MSc. Phan Thi Phuong
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Practices: 30 hours (in lab) Private study: 90 hours (self-study)
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Elementary Financial Mathematics (MTH10201) • Measure Theory and Probability (MTH10401) • Ordinary Differential Equations (MTH00015)

Module objectives/intended learning outcomes	Equip students with advanced knowledge of finance and financial mathematics for discrete stochastic and continuous models.
Content	Covers the basic theory of market derivatives, forwards, options, and swaps.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Thanh, D. N., Hong, P. T. T., Trong, D. D. (2011). *Advanced Financial Mathematics* (in Vietnamese). HCMUS Press. 2. Chance Don M. (1989). *An introduction to options and futures*. Dryden Press. 3. R.L.McDonald (2013). *Derivatives Markets*, 3rd edition, Pearson.

4.2. ELECTIVE COURSES

Quantitative Microeconomics - MTH10214

Module designation	Quantitative Microeconomics
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Phan Thi Phuong
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, do exercises, group work

Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study).
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Algebra A2 (MTH10441)
Module objectives/intended learning outcomes	Equip students with the basic knowledge of finance, financial market operations and financial mathematics for discrete non-random models.
Content	Including the theory of supply and demand model, inflationary, competitive market, and the state policy.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. *Microeconomics*, Tran Nguyen Ngoc Anh Thu, UFM Publishing 2021 (in Vietnamese). 2. *Intermediate Microeconomics*, Hal R.Varian, W.W.Norton & Company Publishing, 2014. 3. *Microeconomics*, Michael Parkin, Pearson Publishing 2022.

Corporate finance - MTH10217

Module designation	Corporate finance
Semester(s) in which the module is taught	7th semester

Person responsible for the module	MSc. Nguyen Huu Toan MSc. Dinh Ngoc Tin
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, exercises
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures 45 hours (in class) Practice: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Capital and capital resources of the enterprise. Cost of capital and capital investment decisions of enterprises. Securities valuation and business valuation. Knowledge: Understand the three basic issues of corporate finance, including which fields to invest in, how to properly mobilize funding sources, and how the business will have to manage daily financial activities. Master the concept of working capital, determine the structure of mobilized capital in a way that ensures minimizing mobilization costs, increasing value for the business while still controlling risks for the business. Skills & competencies: Investment planning (Capital Budgeting). The ability to discover investment opportunities that are likely to generate income that exceeds the cost of financing the implementation of that investment. Determine the structure of mobilized capital to ensure minimizing mobilization costs, increasing value for the business, while still controlling risks for the business.

Content	This module includes the following topics: 1. Introduction to corporate finance. 2. Discounted cash flow technique. 3. Net present value and investment project appraisal methods. 4. Investment decision. 5. Interest rates and bond pricing. 6. Stock valuation. 7. Risk and return: Lessons from the past. 8. Risk and return: Capital asset pricing model (CAPM)
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Main textbook: 1. Stephen A. Ross, *Fundamentals of Corporate Finance*, 13th, McGraw-Hill Education, 2022. References: 1. Allen, Franklin; Brealey, Richard A.; Myers, Stewart C, *Principles of Corporate Finance*, 13th, McGraw-Hill Education, 2020 2. Van Der Post, Hayden, *Python for Corporate Finance: Strategic Financial Modelling: A Comprehensive guide to Financial modelling with Python*, Reactive Publishing, 2023 3. Richard Pike and Bill Neale, *Corporate Finance And Investment Decisions & Strategies,* 5th, Pearson Education, 2006.

Financial Analysis - MTH10219

Module designation	Financial Analysis
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Nguyen Huu Toan MSc. Dinh Ngoc Tin

Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture, exercise
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 75 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Financial reports. Analyze financial statements, treasury, financing activities, investment activities, income, and net cash flow. Knowledge: Provides students with basic knowledge related to the establishment and analysis of financial reports such as balance sheets, business performance reports, and cash flow statements. Skills & competencies: cognitive and practical abilities to use this knowledge in presenting and analyzing data.
Content	This module includes the following topics: 1. Overview of financial statement analysis. 2. Introducing financial reports. 3. Analyze financial ratios. 4. fund analysis and cash flow reports. 5. Analyze sponsorship activities. 6. Analyze investment activities 7. Income analysis 8. Cash flow analysis
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.

Reading list	Main textbook: 1. Weygandt, Kimmel, Kieso, *Accounting Principles*, 13th, Wiley, 2018 References: 1. John Wild, *Financial Accounting Fundamentals*, 8th, McGraw Hill, 2021 2. Thomas R. Robinson; Michael A. Broihahn; *Elaine Henry, International financial statement analysis workbook*, 4th, CFA Institute investment series, 2020. 3. Hayden Van Der Post, *Quantitative Finance with Python: A Deep Dive into*, Reactive Publishing, 2023.

Basic Actuarial Mathematics - MTH10220

Module designation	Basic Actuarial Mathematics
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Dr. Tra Quoc Khanh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study).
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: Measure Theory and Probability (MTH10401)

	Corporate Finance (MTH10217)
Module objectives/intended learning outcomes	General objectives: To gain basic concepts/knowledge on (long-term) actuarial mathematics. Specific objectives/course learning outcomes: Knowledge: To explore basic concepts/notation of long-term actuarial mathematics. To explore how the net premium (or pure premium) is calculated for basic life insurance products. Skills: Calculation skills/ self-study skill. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: 1. Introduction to Life Insurance (or Long-term Insurance) 2. Survival Probabilities 3. Life Table 4. Life Insurance 5. Life Annuities 6. Net Premium Calculation 7. Gross Premium Calculation
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Attendance and Quizzes: 10%, Midterm: 30%, Final: 60%
Reading list	1. D. C. M. Dickson, M. R. Hardy and H. R. Waters, *Actuarial mathematics for life contingent risks*, Cambridge University Press, 2009. 2. Newton L. Bowers et al, *Actuarial Mathematics*, Society of Actuaries, 1997. 3. Robin Cunningham et al, *Models for Quantifying risk*, ACTEX publication, 2012.

Advanced Actuarial Mathematics - MTH10221

Module designation	Advanced Actuarial Mathematics
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Dr. Tra Quoc Khanh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lectures, group work, small group solving exercises
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 hours (in class) Exercises: 30 hours (in class) Private study: 90 hours (self-study)
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: Basic Actuarial Mathematics (MTH10220)
Module objectives/intended learning outcomes	General objectives: To deep dive on the advanced concepts of (long-term) actuarial mathematics. Specific objectives/course learning outcomes: Knowledge: To explore advanced concepts of long-term actuarial mathematics: Reserve, Multiple Decrements model, Multiple Lives model, Multiple States model, Universal Life Insurance, Pension Mathematics. Skills: Calculation skills/ self-study skill. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: Reserve Multiple Decrements Model

	Multiple Lives Model Multiple States Model Universal Life Insurance Pension Mathematics
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Attendance and Quizzes: 10% Midterm: 30%, Final: 60%
Reading list	1. D. C. M. Dickson, M. R. Hardy and H. R. Waters, *Actuarial mathematics for life contingent risks*, Cambridge University Press, 2009. 2. Newton L. Bowers et al, *Actuarial Mathematics*, Society of Actuaries, 1997. 3. Robin Cunningham et al, *Models for Quantifying risk*, ACTEX publication, 2012.

5. SPECIALIZATION IN OPTIMIZATION

5.1. COMPULSORY COURSES

Operations Research - MTH10446

Module designation	Operations Research
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Nguyen Van Thuy
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	Required prerequisites: • Linear Programming (MTH10449)
Module objectives/intended learning outcomes	Provide students with knowledge and applications of linear programming problems in the form of graphs, networks, and some practical applications.
Content	Basic concepts of graphs and trees.

	Algorithms of network problems. Network diagram methods. Concepts of transport problems and algorithms.
Examination forms	Mid-term and Final exam: Written exam
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Phan Quoc Khanh, *Operations Research* (in Vietnamese), Vietnam Education Publishing House, 2004. 2. Phan Quoc Khanh, Tran Hue Nuong, *Linear Programming* (in Vietnamese), Vietnam Education Publishing House, 2003. 3. Frederick Hillier, Gerald Lieberman (2021). *Introduction to Operations Research*, 11th ed. Mc Graw Hill.

Nonlinear programming - MTH10447

Module designation	Nonlinear Programming
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Assoc. Prof. Nguyen Le Hoang Anh
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, do exercises, group work

Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 credits (6 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: • Linear Programming (MTH10449)
Module objectives/intended learning outcomes	Provide students with basic knowledge about the nonlinear programming and optimality conditions of this problem.
Content	Concepts of differentiability in normed spaces. Important theorems in nonlinear analysis: implicit functions, inverse functions, Hahn-Banach, convex set separation theorem, open mapping theorem. Minimum existence theorems Optimality conditions, necessary conditions, sufficient conditions, Lagrange multiplier rules. Duality theorem, saddle point.
Examination forms	Mid-term and Final exam: Written exam
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Johannes Jahn, *Introduction to the Theory of Nonlinear Optimization*, 2007. 2. Patriksson M, *Nonlinear programming and variational inequality problems: a unified approach*, Springer, 2013. 3. William P. Fox. (2021). *Nonlinear Optimization: Models and Applications*. CRC Press.

Optimization Models in Economics - MTH10615

Module designation	Optimization Models in Economics
Semester(s) in which the module is taught	5th semester
Person responsible for the module	MSc. Cao Nghi Thuc
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	150 hours Contact hours: Lectures: 30 lecture hours + 30 exercise hours (in class) Private study: 90 hours (self-study)
Credit points	3 credits (5 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: • Linear programming (MTH10449)
Module objectives/intended learning outcomes	Provides students with the basic concepts of mathematical modeling in economics.
Content	Closed and open Leontief model, direct and indirect costs Labor theory of value, substitution theorem. Linear optimal model, producible set, efficient production, production constraints, consumption.

	Nonlinear optimization model, modern theory of demand, extended theory of production. Equilibrium in the market economy, budget constraints, Walras-Wald model, Arrow-Debreu-Mckenzie model.
Examination forms	Mid-term and Final exam: Written exam
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Dowling Edward T, *Introduction to Mathematical Economics*, Schaum's Outline Series, 2012. 2. Hoang Dinh Tuan, *Theory of Mathematical Models in Economics* (in Vietnamese), National Economics University, 2007. 3. Mikulás Luptácik (2010). *Mathematical Optimization and Economic Analysis*. Springer.

Linear programming - MTH10449

Module designation	Linear Programming
Semester(s) in which the module is taught	5th semester
Person responsible for the module	Assoc. Prof. Nguyen Le Hoang Anh MSc. Nguyen Manh Truong Giang
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, do exercises, group work

Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 lectures hours + 30 exercises hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Provide students with the tools to solve the linear programming problem, distinguish the types of problems, the original monomorphic, dual, and synthetic methods. How to build models from real problems.
Content	Objective function, constraint, sign constraint, vocabulary, basis solution, extreme point, primal simplex method, duality.
Examination forms	Mid-term and Final exam: Written exam
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Phan Quoc Khanh, Tran Hue Nuong, *Linear Programming* (in Vietnamese), Vietnam Education Publishing House, 2003. 2. Robert J. Vanderbei, *Linear Programming: Foundations and Extensions*, Springer, 2020. 3. Jiří Matoušek , Bernd Gärtner (2007). *Understanding and Using Linear Programming*. Springer.

Optimization Algorithms - MTH10450

Module designation	Optimization Algorithms

Semester(s) in which the module is taught	6th semester
Person responsible for the module	Assoc. Prof. Vo Si Trong Long
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, lab work, group work
Workload (incl. contact hours, self-study hours)	165 hours Contact hours: Lectures: 45 lectures hours + 30 lab works hours (in class) Private study: 90 hours (self-study)
Credit points	4 credits (6.5 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Provide students with knowledge and methods to solve constrained and unconstrained optimization problems.
Content	Basic properties and concepts of algorithms, Direct/Indirect line search method for unconstrained optimization problem. Deep reduction method, gradient method and conjugate gradient method. Newton and quasi-Newton methods. The least squares problem. Kuhn-Tucker Optimality Conditions.

Examination forms	Mid-term and Final exam: Written exam.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. Nocedal J, Wright SJ: *Numerical optimization*. Springer, New York, 1999 2. Polak Elijah, *Optimization: algorithms and consistent approximations*, Springer, 1997. 3. Alaa Khamis (2024). *Optimization Algorithms*. Manning Publications.

Introduction to Convex Analysis and Convex Programming - MTH10543

Module designation	Introduction to Convex Analysis and Convex Programming
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Assoc. Prof. Vo Si Trong Long
Language	Vietnamese
Relation to curriculum	Compulsory for Specialization in Optimization, AM programme. Elective for others programme.
Teaching methods	Lecture, do exercises, group work
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 60 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6 ECTS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	Provide students with knowledge about convex analysis.
Content	Convex set, affine set, Caratheodory's Theorem. Convex function, quasi-convex function, lower and upper semicontinuous, continuous, Lipschitz continuous, Karamardian theorem, local minima. Hahn-Banach theorem, separation form. Convex programming. Optimal conditions. Duality theorem.
Examination forms	Mid-term and Final exam: Written exam
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	1. R.T. Rockafellar, *Convex Analysis*, Princeton University Press, Princeton, New Jersey, 1970 2. Boris S. Mordukhovich and Nguyen Mau Nam, *An Easy Path to Convex Analysis and Applications*, 2nd edn. Springer, Cham, Switzerland, 2023. 3. Steven G. Krantz (2015). *Convex Analysis.* CRC Press.

5.2. ELECTIVE COURSES

Game Theory - MTH10541

Module designation	Game Theory
Semester(s) in which the module is taught	6th semester

Person responsible for the module	MSc. Nguyen Van Thuy
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: • Operations Research (MTH10446)
Module objectives/intended learning outcomes	Introduces the basics of game theory.
Content	This module includes the following topics: 1. Some basis definitions. 2. Minimax Theorem 3. 2-person zero-sum games 4. symmetric zero-sum game 5. Indeterminate games, countable strategies, convex-concave games. 6. Non-zero sum game
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Class attendance: at least 70%. Final score is greater or equal to 5.0/10.0
Reading list	1. G. Owen, *Game Theory*, Academic Press Inc, 1995. 2. Bazaraa Mokhtar S, Jarvis John J., Sherali Hanif D., *Linear programming and network flows*. Wiley, 1990. 3. Steven Tadelis (2013). *Game Theory: An Introduction*. Princeton University Press.

Optimal Control - MTH10545

Module designation	Optimal Control
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Cao Nghi Thuc
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lecture
Workload (incl. contact hours, self-study hours)	180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 Credits (6.0 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: • Nonlinear Programming (MTH10447)
Module objectives/intended learning outcomes	This course provides fundamenta7l knowledge on the Optimal control problem, Pontryagin's maximum principle, Existence and continuity properties of optimal controls.
Content	This module includes the following topics: Introduce the optimal control problem. Conditions for the existence of solutions to the optimal control problem. Pontryagin's maximum principle to solve many kinds of the optimal control problem.
Examination forms	Midterm and final exam: written exams.
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.

Reading list	1. Vu Ngoc Phat, *Introduction to Mathematical Control Theory* (in Vietnamese), Vietnam National University Press, Hanoi, 2001. 2. Johannes Jahn, *Vector Optimization*, Springer, 2004. 3. Gamkrelidze, R. *Principles of optimal control theory.* Vol. 7. Springer Science & Business Media, 2013.

Multi-objective Optimization - MTH10553

Module designation	Multi-objective Optimization
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Manh Truong Giang
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Project, seminar, group work
Workload (incl. contact hours, self-study hours)	150 Hours Contact hours: Lectures: 60 hours (in class) Private study: 90 hours (self-study)
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	Recommended prerequisites: Linear programming (MTH10449)
Module objectives/intended learning outcomes	This course provides fundamental knowledge about Multi objective problems and algorithms related to this class of problems. Knowledge: get familiar with basic concepts in the theory of multi-objective optimization problem model, concept of solution and corresponding algorithm. Skills & competences: Establish multi-objective optimization models for practical problems. Describe multi-objective optimization solution structures. Appreciate multi-

	objective optimization methods. Solve bicriteria linear optimization problems. Attitude: Attend all classroom sessions, and discussions outside of class time.
Content	This module includes the following topics: Solution concepts and some properties of solutions Duality Multi-objective linear programming Multi-objective programming
Examination forms	Midterm and final exam: written exams
Study and examination requirements	Class attendance: at least 70%. Final score is greater or equal to 5.0/10.0
Reading list	1. M. Ehrgott, *Multicriteria Optimization*, Springer, 2005. 2. D. T. Luc, *Multiobjective Linear Programming*, Springer, 2014. 3. Yann Collette, Patrick Siarry (2004). *Multiobjective Optimization: Principles and Case Studies*. Springer.

Seminar on Optimization - MTH10616

Module designation	Seminar on Optimization
Semester(s) in which the module is taught	7th semesters
Person responsible for the module	Assoc. Prof. Nguyen Le Hoang Anh
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	project, seminar, group work

Workload (incl. contact hours, self-study hours)	Total workload:180 Hours Contact hours: Lectures: 60 hours (in class) Private study: 120 hours (self-study)
Credit points	4 credits (6 ECTS)
Required and recommended prerequisites for joining the module	This course is only for students who meet the following conditions: Overall grade: at least 6.5/10 at the time of registration. Graduate with a major in Optimization.
Module objectives/intended learning outcomes	Learn more about the Optimization major and prepare for your graduation thesis.
Content	Students will work directly with instructors on a topic. Topic content can be: Issues that have not been mentioned in the subjects of the Optimization major; Problems that arise in Optimization subjects but have not been resolved. The latest results from around the world and upcoming results from professors, lecturers, and graduate students of the Department of Optimization and Systems.
Examination forms	Mid-term and Final exam: report (or written exam, if necessary)
Study and examination requirements	Class attendance: at least 70%. Overall grade: minimum 5.0/10.0.
Reading list	Updated from the latest scientific information depending on the discussion content and provided by the instructor.

6. GRADUATING WORKS

Internship - MTH10549

Module designation	Internship
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Project, seminar, group work
Workload (incl. contact hours, self-study hours)	Total workload: 180 hours Contact hours (lectures, exercises): 90 hours Private study including examination preparation, specified in hours: 90 hours
Credit points	4 Credits (6 ECTS)
Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: Apply knowledge, skills in mathematics/applied mathematics/computer science to practice with real-life problems in company, industry. Self-orient in the desired professional field. Understand the impact of mathematics and computer science in a global, economic, environmental, and societal context. Conduct work responsibly in cooperation with others. Write and present a report in an adequate format.
Content	Attendance: Students should attend 100%. Attendance will be regulated and checked. Individual Assignments: Tasks are assigned by the industrial company.

	Internship: At a company in the field corresponding to the registered major.
Examination forms	Internship report, Oral presentation
Study and examination requirements	Internship certificate issued by the company: 50% Written report and presentation graded by the faculty supervisor: 50%
Reading list	Depending on the assigned tasks and instructors.

Graduation Thesis - MTH10595

Module designation	Graduation Thesis
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Assigned lecturers of the Faculty of Mathematics and Computer Science, University of Science, VNU-HCM
Language	Vietnamese
Relation to curriculum	Compulsory
Teaching methods	Project, seminar
Workload (incl. contact hours, self-study hours)	Total workload: 300 hours Contact hours: 30 hours Private study including examination preparation, specified in hours: 270 hours
Credit points	10 credits (20 ECTS)
Required and recommended prerequisites for joining the module	Successfully completed 96/131 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: Point out the research problems Apply fundamental knowledge and theories to analyze the research problem; develop a research framework (model if applicable); and design a solution for the problem

	Use appropriate techniques for solutions Demonstrate the scientific contribution and practical relevance of the research carried out Provide a consistent, well-structured Bachelor Thesis Plan and manage learning process Present and defend results of the thesis at the Students' Bachelor Thesis conference.
Content	Various topics
Examination forms	Graduation report
Study and examination requirements	Final defense: 100%
Reading list	Diverse and depends on the supervisors

Surveys of Mathematics, Computer Science, and specializations - MTH10617

Module designation	Surveys of Mathematics, Computer Science, and specializations.
Semester(s) in which the module is taught	4th semester
Person responsible for the module	The representative of each major.
Language	Vietnamese
Relation to curriculum	Elective
Teaching methods	Lectures, group work
Workload (incl. contact hours, self-study hours)	Total workload: 90 Hours Contact hours: Lectures: 30 hours (in class) Private study: 60 hours (self-study)
Credit points	2 Credits (3 ECTS)

Required and recommended prerequisites for joining the module	None
Module objectives/intended learning outcomes	General Objective: The course aims to provide students with a broad understanding of mathematics, computer science, and related majors in the training program. This foundation helps students orient themselves and make informed decisions when selecting their major. Specific Objectives / Course Learning Outcomes: Knowledge: Students will gain an overview of mathematics, computer science, and related majors, along with their applications and career opportunities. Skills: Gaining a general understanding of a field. Attitude and Diligence: Seriousness, diligence, and a scientific approach.
Content	The course consists of lectures and activities conducted by multiple instructors, providing an overview of mathematics and computer science, their history worldwide, and the current state in Vietnam. It also introduces various majors, including their focus, research scope, applications, and career opportunities. Students engage actively, provide feedback, and undergo assessments.
Examination forms	written exams
Study and examination requirements	Tests - process score: 70% Final exam: 30 %
Reading list	1. *History of Mathematics* (in Vietnamese), Nguyen Phu Loc, Vietnam Education Publishing House, 2008. 2. *The History of the Development of Information Technology and its Organizational and Societal Impact*. M.C. Mr, 2013. 3. Lecture notes.